Introduction to DBMS, Basic Database Terminology

Here is a **simple and clear Introduction to DBMS** and **Basic Database Terminology —** perfect for exams, viva, or quick revision.

📘 Introduction to DBMS

A **Database Management System (DBMS)** is software used to **create, store, manage, and retrieve data** efficiently.

Instead of storing data in files manually, DBMS helps maintain large amounts of data with:

- Accuracy
- Security
- Fast access
- Easy updates
- Reduced redundancy

Why DBMS?

Before DBMS, data was stored in "File-based Systems," which had problems like:

- Data redundancy (duplicate data)
- Data inconsistency
- Difficulty in searching and updating
- Poor data security
- Lack of backup and recovery

A DBMS solves these problems by organizing data scientifically.

🎯 Functions of DBMS

- **Data Storage & Management**
- **Data Retrieval (Queries)**
- **Data Security & Authorization**
- **Backup & Recovery**

- **Concurrency Control** (multiple users at the same time)
- **Data Integrity** (accuracy and correctness)

🏛️ Types of DBMS

1. **Relational DBMS (RDBMS)**
   - Uses tables (rows & columns)
   - Example: MySQL, Oracle, PostgreSQL, SQL Server
2. **Hierarchical DBMS**
   - Tree-like structure
   - Example: IBM IMS
3. **Network DBMS**
   - Complex many-to-many relationships
4. **Object-Oriented DBMS (OODBMS)**
   - Stores objects (used in advanced systems)

📌 Basic Database Terminology

Here are the most important terms:

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1. Database

A **collection of related data** stored in an organized way.

Example:

A college database contains tables like Students, Courses, Fees, Marks, etc.

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2. DBMS

Software that interacts with the database.

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3. Table (Relation)

A structure in RDBMS to store data in **rows and columns**.

Example: Student table with columns (ID, Name, Age).

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4. Tuple (Row/Record)

A **single row** in a table.

Example:

```
101 | Yuvraj | 22
```

---

5. Attribute (Column/Field)

A **column** in a table.

Example: ID, Name, Age are attributes.

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6. Domain

The **set of valid values** an attribute can take.

Example: Age domain = 1 to 120.

---

7. Primary Key

A unique identifier for each record.

Example: RollNo, AadharNo, Email (unique per student).

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8. Candidate Key

All possible attributes that can be primary keys.

Example:
In Student table → (RollNo, Email) both are unique → both are candidate keys.

9. Super Key

Any set of attributes that can uniquely identify a record.

Super key = Candidate Key + Extra attributes.

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10. Foreign Key

A key that refers to a primary key in another table.

Example:

`Student.CourseID → Course.ID`

---

11. Composite Key

Primary key made of **two or more attributes**.

Example:

`(StudentID, CourseID)` in an enrollment table.

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12. Schema

The **structure** of the database.

Example:

Tables, attributes, types, constraints—all part of schema.

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13. Instance

Data **at a particular moment** (snapshot).

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14. Metadata

Data **about data** (schema, definitions).

---

15. Constraints

Rules applied to data:

- NOT NULL
- UNIQUE
- CHECK
- DEFAULT
- PRIMARY KEY
- FOREIGN KEY

📘 Summary Table

Term	Meaning
Database	Organized collection of data
DBMS	Software for managing databases
Table	Collection of rows and columns
Tuple	A row/record
Attribute	A column
Domain	Allowed values of a column
Primary Key	Unique identifier
Foreign Key	References primary key of another table
Schema	Overall structure
Instance	Data at a moment
Metadata	Data about data

If you want, I can also provide:

✅ **Notes in PDF format**
✅ **MCQs with answers**
✅ **Short/long exam questions**
✅ **ER Diagram explanation**

Just tell me what you need!

Types of DBMS: Relational, Object Relational and NoSQL Databases

Here is a **clear, exam-friendly explanation** of the three major types of DBMS: **Relational, Object-Relational, and NoSQL Databases**.

📘 1. Relational Database Management System (RDBMS)

Definition

An RDBMS stores data in the form of **tables (relations)** consisting of **rows** and **columns**. It follows **E. F. Codd's 12 rules** and uses **Structured Query Language (SQL)**.

Data Model

- **Tables** (Relations)
- **Rows** (Tuples)
- **Columns** (Attributes)
- **Keys** (Primary, Foreign, Candidate)

Characteristics

- Structured data
- Follows **ACID** properties (Atomicity, Consistency, Isolation, Durability)
- Strong consistency & integrity
- Uses **SQL** for queries

Examples

- MySQL
- PostgreSQL
- Oracle
- MS SQL Server
- MariaDB

Advantages

- Simple and structured
- Ensures data accuracy and consistency
- Strong security
- Supports complex queries

Limitations

- Difficult to handle **unstructured** or **semi-structured** data
- Not easily scalable for big data (vertical scaling only)

📘 2. Object-Relational Database Management System (ORDBMS)

Definition

An ORDBMS is a **hybrid model** that combines:

- Relational database features
- Object-oriented programming concepts (objects, classes, inheritance, methods)

It extends RDBMS to store **complex data types**.

Data Model

- Tables (like RDBMS)
- **User-defined types (UDT)**
- Objects
- Methods (functions associated with objects)
- Inheritance

Characteristics

- Supports complex structures like:
  - Images
  - Audio/video
  - Large objects (LOBs)
  - Arrays, JSON, XML
- Bridges the gap between object-oriented applications and relational storage.

Examples

- PostgreSQL (most popular ORDBMS)
- Oracle (supports object types)
- Informix
- DB2

Advantages

- Can handle complex data
- Better for multimedia, scientific, and engineering applications
- Supports object features inside SQL

Limitations

- More complex to implement
- Higher cost compared to simple RDBMS
- Performance overhead due to object features

📘 3. NoSQL Databases

Definition

NoSQL means **Not Only SQL**.
These databases handle **large-scale, distributed, unstructured or semi-structured data**.

They do not use traditional tables; instead, they use flexible models.

Why NoSQL?

- Big data
- Real-time applications
- Horizontal scaling (scale-out across many machines)

Types of NoSQL Databases

1. **Document Stores**
   - JSON/XML documents
   - Example: MongoDB, CouchDB
2. **Key-Value Stores**
   - Simple key-value pairs
   - Example: Redis, DynamoDB
3. **Column Family Stores**
   - Data stored in columns instead of rows
   - Example: Cassandra, HBase
4. **Graph Databases**
   - Nodes & edges (relationships)
   - Example: Neo4j, Amazon Neptune

Characteristics

- Schema-less
- Supports unstructured data
- Highly scalable and distributed
- Eventual consistency (not always ACID)

Examples

- MongoDB
- Cassandra
- Redis
- Neo4j
- DynamoDB

Advantages

- High performance
- Easy to scale
- Handles massive amounts of data
- Flexible (no predefined schema)

Limitations

- Less consistency (uses BASE model)
- No standard query language

- Limited multi-row transactions in some systems

📘 Summary Table

Type	Model	Suited For	Examples
RDBMS	Tables (Rows/Columns)	Structured data, transactions	MySQL, PostgreSQL
ORDBMS	Tables + Objects	Complex data types, multimedia	PostgreSQL, Oracle
NoSQL	Documents, Key-Value, Graph, Column	Big data, distributed systems	MongoDB, Cassandra, Redis

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If you want, I can also provide:

📌 **Short answers + 2 mark notes**
📌 **Diagram comparing all three**
📌 **PDF notes**
📌 **MCQs with answers**

Just tell me!

Introduction to MySQL, MySQL Clients (Monitor, Shell, Workbench)

Here is a **clear, exam-oriented explanation** on **Introduction to MySQL** and **MySQL Clients (Monitor, Shell, Workbench)** — easy for notes, viva, or theory exams.

📘 Introduction to MySQL

What is MySQL?

MySQL is an **open-source Relational Database Management System (RDBMS)** developed by Oracle.
It uses **SQL (Structured Query Language)** to create, manage, and retrieve data.

MySQL is one of the most widely used databases in:

- Web applications
- Enterprise software
- Cloud-based systems
- Big websites (YouTube, Facebook – earlier versions)

Key Features of MySQL

- Open-source and free (community edition)
- Uses SQL for querying
- Fast, reliable, scalable
- Supports ACID transactions
- Multi-user access
- Works well with PHP, Java, Python, Node.js, etc.
- Supports replication, clustering, stored procedures, triggers

Common Uses

- Web applications
- E-commerce websites
- Banking & finance
- College/Office management systems
- Data warehousing

📘 MySQL Clients

A **MySQL Client** is any software/tool used to **connect to the MySQL Server** and execute SQL commands.

MySQL supports three main types of clients:

1. **MySQL Monitor (mysql Client)**
2. **MySQL Shell**
3. **MySQL Workbench**

1️⃣ MySQL Monitor (mysql Client) – Command Line Tool

What is MySQL Monitor?

The **MySQL Monitor** is the traditional **command-line client** that comes with MySQL.
It is accessed by typing:

```css
mysql -u root -p
```

Features

- Simple text-based interface
- Allows executing SQL queries directly
- Lightweight and fast
- Good for basic database operations
- Supports running scripts (`.sql` files)

Use Cases

- Creating and managing databases
- Running SQL queries
- Exporting and importing data
- Ideal for servers and terminal-based environments

2️⃣ MySQL Shell (mysqlsh)

What is MySQL Shell?

MySQL Shell is an **advanced, modern client** that supports:

- **SQL mode**
- **JavaScript mode**

- **Python mode**

You open it using:

```nginx
mysqlsh
```

Features

- Supports SQL + scripting (Python, JavaScript)
- Advanced diagnostics
- Supports MySQL X DevAPI
- Allows working with both **Relational tables** and **Document Store**
- Useful for DevOps and automation
- JSON data handling is easier

Use Cases

- Managing MySQL servers and clusters
- Automating tasks using Python/JS
- Working with MySQL Document Store
- Advanced administration

3️⃣ MySQL Workbench – GUI Tool

What is MySQL Workbench?

MySQL Workbench is a **Graphical User Interface (GUI)** tool to interact with MySQL visually.

Features

- Visual query editor
- ER diagram designing (Modeling)
- Database administration
- View, add, update, delete data easily
- Server configuration tools
- Backup & restore
- User management (create roles/users)

Use Cases

- Beginners learning SQL
- Developers designing databases
- Admins managing servers visually
- Generating ER Diagrams automatically
- Running complex queries without CLI

📌 Difference Between Clients

Feature	MySQL Monitor	MySQL Shell	MySQL Workbench
Interface	Command-line	Command-line (Advanced)	Graphical (GUI)
Modes	SQL only	SQL, Python, JavaScript	SQL (GUI-based)
Best For	Basic operations	Automation & DevOps	Database design & admin
Document Store Support	No	Yes	Yes (limited)
Ease of Use	Hard for beginners	Medium	Very easy

📘 Summary

- **MySQL** is a popular open-source RDBMS.
- **MySQL Monitor** → Basic command-line SQL client.
- **MySQL Shell** → Advanced CLI with scripting support (Python/JS).
- **MySQL Workbench** → GUI tool for designing, managing, and querying databases.

If you want, I can also prepare:

📌 **Short notes (2–3 marks)**
📌 **Long answer (10 marks)**
📌 **PDF notes**
📌 **Difference tables**
📌 **MySQL architecture diagram**

Just tell me!

Data Models (Conceptual, Logical, Physical)

Below is an **easy, exam-focused explanation** of **Data Models**—Conceptual, Logical, and Physical.
Perfect for 2-mark, 4-mark, and long-answer questions.

📘 Data Models in DBMS

A **Data Model** is a way of describing how data is stored, organized, and used in a database system.

Data models are used during **database design** to convert real-world requirements into a structured database.

There are **3 main levels** of data modeling:

1. **Conceptual Data Model**
2. **Logical Data Model**
3. **Physical Data Model**

These three levels are part of the **3-Level ANSI/SPARC Architecture**.

1️⃣ Conceptual Data Model (High Level / Business Level)

Definition

A **Conceptual Data Model** represents the **overall structure** of the entire database system at a **high level**.
It shows *what* data is stored and *how* different data entities are related.

Purpose

- Understand business requirements
- Communicate with stakeholders (non-technical people)

Characteristics

- High-level design
- No technical details
- No primary keys, foreign keys, or data types
- Focuses on:
  - **Entities**
  - **Attributes**
  - **Relationships**

Example

ER Diagram of a college system:

- Entities: Student, Course, Faculty
- Relationships: Student enrolls in Course, Faculty teaches Course

Who uses it?

- Business Analysts
- Project Managers
- Designers

2️⃣ Logical Data Model (LDM) – Detailed Database Structure

Definition

A **Logical Data Model** provides a detailed representation of data elements and relationships without considering the physical database.

It describes *how* data will be logically organized.

Characteristics

- More detailed than conceptual model
- Includes:
  - Primary keys
  - Foreign keys
  - Attributes with data types
  - Normalization
  - Constraints
- Still **no physical storage details**

Purpose

- Bridge between conceptual model and physical model
- Logical design for RDBMS

Example

From above:

Student (StudentID PK, Name, Age, CourseID FK)

Course (CourseID PK, CourseName)

Who uses it?

- Database Designers
- System Analysts
- Developers

3️⃣ Physical Data Model (PDM) – Actual Implementation Level

Definition

A **Physical Data Model** shows how the database will be physically stored in the system.

It is the **actual implementation** of the logical model on a database server (e.g., MySQL, Oracle).

Characteristics

- Actual table creation scripts
- Indexes
- Partitioning
- Data files and storage parameters
- Physical constraints
- Performance optimization details
- Hardware/storage considerations

Purpose

- Implement the database
- Improve performance
- Allocate storage resources

Example

SQL Schema:

sql Copy code

```sql
CREATE TABLE Student (
    StudentID INT PRIMARY KEY,
    Name VARCHAR(50),
    Age INT,
    CourseID INT,
    INDEX idx_course (CourseID),
    FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);
```

Who uses it?

- Database Administrators (DBAs)
- Backend Developers
- System Architects

📌 Comparison Table

Feature	Conceptual	Logical	Physical
Level	High-level	Mid-level	Low-level
Focus	Entities, relationships	Attributes, keys, constraints	Storage, indexes, files
Audience	Business users	Designers/Developers	DBAs
Includes PK/FK?	❌ No	✅ Yes	✅ Yes
Includes data types?	❌ No	✅ Yes	✅ Yes + engine details
Technology dependent?	❌ No	❌ No	✅ Yes (MySQL, Oracle, etc.)

📘 Summary

- **Conceptual Model** → High-level business view (entities & relationships).
- **Logical Model** → Detailed structure with attributes, keys, normalization.
- **Physical Model** → Actual database implementation (SQL, indexes, storage).

If you want, I can also prepare:

📌 **Short notes (2–3 marks)**
📌 **ER diagram example**
📌 **PDF notes**
📌 **MCQs for practice**

Just tell me!

Database Design, Entity-Relationship Diagram (ERD)

Here is a **simple, clear, and exam-oriented explanation** of **Database Design** and **Entity–Relationship Diagram (ERD) —** perfect for theory exams, viva, and assignments.

📘 Database Design

What is Database Design?

Database Design is the process of **planning, structuring, and organizing data** to build an efficient, reliable, and consistent database.

It converts real-world requirements into a structured database that supports:

- Fast retrieval
- Easy modification
- Data accuracy
- Minimal redundancy
- Data integrity

📌 Steps in Database Design

Database design is typically done in **six major steps**:

1️⃣ Requirements Analysis

- Collect user and system requirements
- Identify what data needs to be stored
- Understand business rules

Example:

For a college system → Students, Courses, Faculty, Fees, Exams.

2️⃣ Conceptual Design

- Create a **Conceptual Data Model**

- Build an **ER Diagram (Entity-Relationship Diagram)**
- Identify:
  - Entities
  - Attributes
  - Relationships

---

3 Logical Design

- Convert the conceptual model into a **Logical Schema**
- Decide:
  - Tables
  - Columns (attributes)
  - Primary & Foreign Keys
  - Constraints
  - Normalization

---

4 Physical Design

- Define how data will be stored on disk
- Create actual **SQL tables**
- Add:
  - Indexes
  - Partitions
  - Storage engines
  - Performance settings

---

5 Implementation

- Use SQL (e.g., MySQL, Oracle, PostgreSQL)
- Create tables, views, triggers, stored procedures

---

6 Testing & Maintenance

- Test for performance & security

- Optimize queries
- Update design for new requirements

📘 Entity-Relationship Diagram (ERD)

What is an ERD?

An **Entity-Relationship Diagram (ERD)** is a graphical representation of a database at the **conceptual level**.

It shows:

- **Entities** (things in real world)
- **Attributes** (properties of entities)
- **Relationships** (how entities are connected)

ERD is the most important tool in database design.

📌 ERD Components

1️⃣ Entity

An **Entity** is an object in the real world that has an independent existence.

Examples:

- Student
- Course
- Employee
- Customer

Representation:

📦 Rectangle

2️⃣ Attribute

An **Attribute** is a property of an entity.

Examples:

- Student: RollNo, Name, Age
- Course: CourseID, CourseName

Types of Attributes

- **Simple** (Name, Age)
- **Composite** (FullAddress = Street + City + PIN)
- **Derived** (Age calculated from DOB)
- **Multivalued** (PhoneNumber, Email)

Representation:

🔘 Oval shape

3️⃣ Key Attribute

Uniquely identifies an entity instance (Primary Key).

Example: StudentID, EmployeeID.

Representation:

Oval with **underline**

4️⃣ Relationship

Defines how entities are related.

Examples:

- Student **enrolls in** Course
- Employee **works for** Department

Representation:

◆ Diamond shape

📘 Types of Relationships in ERD

1️⃣ One-to-One (1:1)

One entity relates to one entity.

Example:

Employee — has — AadhaarCard

2️⃣ One-to-Many (1:N)

One entity relates to many entities.

Example:

Department — has — Employees

3️⃣ Many-to-Many (M:N)

Many entities relate to many other entities.

Example:

Student — enrolls in — Courses

(Requires a **bridge table** during logical design.)

📘 Example ER Diagram

University Database

Entities:

- Student(StudentID, Name, Email)
- Course(CourseID, CourseName, Credits)
- Faculty(FacultyID, Name, Dept)

Relationships:

- Student **enrolls in** Course (M:N)
- Faculty **teaches** Course (1:N)

ERD (Text Representation):

pgsql Copy code

```
Student (StudentID, Name, Email)
        |
        | enrolls in (M:N)
        |
Course (CourseID, CourseName, Credits)
        |
        | taught by (1:N)
        |
Faculty (FacultyID, Name, Dept)
```

📘 Importance of ERD

- Helps understand system requirements
- Reduces data redundancy
- Improves database consistency
- Acts as blueprint for database design
- Communicates structure to developers & stakeholders

📘 Summary

- **Database Design** is the complete planning process for building a database.
- **ERD** is a visual model representing entities, attributes, and relationships.
- ERD is created during **Conceptual Design** and guides logical & physical design.

If you want, I can also prepare:

📌 **Short notes (2–3 marks)**
📌 **ERD diagrams for College, Hospital, Banking, Library systems**
📌 **MCQs with answers**
📌 **PDF version of these notes**

Just tell me!

Codd's 12 rules for RDBMS

Here is a **clear, concise, exam-friendly explanation** of **Codd's 12 Rules for RDBMS —** perfect for theory, viva, and long answers.

📘 Codd's 12 Rules for RDBMS

E. F. Codd, the father of the relational database model, proposed **12 rules** to define what a **true Relational Database Management System (RDBMS)** should follow.

These rules help ensure consistency, integrity, and relational capability.

Rule 0 – Foundation Rule

A system must use its **relational capabilities** *exclusively* to manage the database.

If a system claims to be RDBMS, it must:

- Store data in tables
- Use relational operations
- Support SQL

Rule 1 – Information Rule

All information in a database must be represented **only in one way**:

📌 **As values in tables (rows and columns).**

No other form is allowed (e.g., files, pointers).

Rule 2 – Guaranteed Access Rule

Every data element (value) must be reachable by:

1. Table name
2. Primary key (row)
3. Attribute name (column)

This ensures **controlled, precise access** to data.

Rule 3 – Systematic Treatment of Null Values

NULL values should be:

- Systematically allowed
- Represent missing/unknown/not applicable data
- Treated uniformly by the system

NULL must not be interpreted as 0 or empty string.

Rule 4 – Dynamic Online Catalog Based on the Relational Model

Metadata (data about data), such as:

- Table definitions
- Attribute names
- Constraints

must be stored in **tables** and accessible using SQL.

Rule 5 – Comprehensive Data Sub-Language Rule

A relational system must support at least one **complete language** (like SQL) with:

- Data definition
- Data manipulation
- Data control
- Transaction control

- View definition
- Integrity constraints

Rule 6 – View Updating Rule

All views that are **theoretically updatable** must be **updatable** through the database system.

Users must be able to insert/update data through views when logically possible.

Rule 7 – High-Level Insert, Update, and Delete Rule

The system must support:

- INSERT
- UPDATE
- DELETE

operations that work on **sets of rows**, not only row-by-row.

Example:
Delete all students with marks < 40.

Rule 8 – Physical Data Independence

Changes in **physical storage** should *not* affect:

- Logical structure
- Application programs

Example:
Changing file location should not break queries.

Rule 9 – Logical Data Independence

Changes in **logical structure** should *not* affect application programs.

Example:
Adding a new column to a table should not break existing queries.

This is harder to achieve than physical independence.

Rule 10 – Integrity Independence

Integrity constraints (like PK, FK, CHECK) must be **stored in the catalog** and **not in application code**.

The database—not the application—should handle:

- Primary keys
- Foreign keys
- Domain constraints
- Business rules

Rule 11 – Distribution Independence

The system must be able to work **unchanged** whether the data is distributed over multiple locations or stored on one machine.

Distributed architecture must be transparent to users.

Rule 12 – Non-Subversion Rule

If the system has low-level access (e.g., record-level operations), it must not **bypass** relational rules.

All operations must obey relational integrity.

📘 Summary Table

Rule No.	Rule Name	Meaning
0	Foundation	Must use only relational model
1	Information	All data stored in tables
2	Guaranteed Access	Access by table, key, and column
3	Null Values	Uniform treatment of NULL
4	Online Catalog	Metadata stored as tables
5	Sub-Language	Must have complete SQL
6	View Updating	Views must be updatable
7	High-Level Update	Set-based operations
8	Physical Independence	Physical changes don't affect logic
9	Logical Independence	Logical changes don't affect apps
10	Integrity	Constraints stored in DB
11	Distribution	Distributed DB transparency
12	Non-Subversion	Low-level ops must follow rules

📘 2–3 Mark Short Answer (Ready to Write in Exam)

Codd's 12 rules define the essential characteristics of a true RDBMS.
They ensure data is stored as tables, accessed using keys, support SQL, enforce integrity, maintain data independence, support updatable views, allow set-based operations, and provide distribution transparency.

If you want, I can also give:

📌 **10-mark long answer**
📌 **MCQs with answers**
📌 **PDF of all notes so far**

Just tell me!

Introduction to SQL, Categories of SQL Commands: DDL, DML, DCL, DTL/TCL

Here is a **clear, concise, and exam-focused explanation** of **Introduction to SQL** and **Categories of SQL Commands (DDL, DML, DCL, DTL/TCL)** — perfect for 2–10 mark questions.

📘 Introduction to SQL

What is SQL?

SQL (Structured Query Language) is a standard language used to **store, manage, and retrieve data** from a relational database.

It is used in almost all RDBMS systems like:

- MySQL
- Oracle
- SQL Server
- PostgreSQL
- MariaDB

Key Features of SQL

- Easy to learn
- Uses English-like syntax
- Standard language across all RDBMS

- Supports complex queries and joins
- Allows transaction management
- Ensures data security

What SQL Can Do?

- Create and modify database objects
- Insert, update, delete data
- Retrieve data using queries
- Control user access
- Manage transactions (COMMIT, ROLLBACK)

📘 Categories of SQL Commands

SQL commands are classified into four main categories:

1. **DDL – Data Definition Language**
2. **DML – Data Manipulation Language**
3. **DCL – Data Control Language**
4. **TCL (or DTL) – Transaction Control Language**

Let's understand each one.

1️⃣ DDL – Data Definition Language

Used to **define or modify database structure** (tables, indexes, views).

Common DDL Commands

Command	Purpose
CREATE	Create table, database, view, index
ALTER	Modify table structure
DROP	Delete table or database
TRUNCATE	Remove all rows quickly (no rollback)

Command	Purpose
RENAME	Rename table

Examples

sql Copy code

```sql
CREATE TABLE Student (
    ID INT PRIMARY KEY,
    Name VARCHAR(50)
);

ALTER TABLE Student ADD Age INT;
```

2️⃣ DML – Data Manipulation Language

Used to manipulate (modify) **data stored inside tables**.

Common DML Commands

Command	Purpose
INSERT	Add new rows
UPDATE	Modify existing rows
DELETE	Remove rows
SELECT	Retrieve data (Read operation)

Examples

sql Copy code

```sql
INSERT INTO Student VALUES (101, 'Yuvraj', 20);

UPDATE Student SET Age = 21 WHERE ID = 101;
```

```sql
DELETE FROM Student WHERE ID = 101;
```

3️⃣ DCL – Data Control Language

Used to **control access** to data in the database.

Common DCL Commands

Command	Purpose
GRANT	Give permission to users
REVOKE	Remove permission from users

Examples

sql Copy code

```sql
GRANT SELECT, INSERT ON Student TO user1;

REVOKE INSERT ON Student FROM user1;
```

4️⃣ TCL (or DTL) – Transaction Control Language

Used to **manage transactions**, ensure data consistency, and handle changes.

A **transaction** is a sequence of operations performed as a single logical unit.

Common TCL Commands

Command	Purpose
COMMIT	Save changes permanently
ROLLBACK	Undo changes
SAVEPOINT	Create a point to roll back to

Command	Purpose
SET TRANSACTION	Set transaction property (read-only, isolation level)

Examples

sql Copy code

```sql
BEGIN;

UPDATE Account SET Balance = Balance - 500 WHERE ID = 1;
UPDATE Account SET Balance = Balance + 500 WHERE ID = 2;

COMMIT;
```

OR, if error:

sql Copy code

```sql
ROLLBACK;
```

📘 Summary Table

Category	Full Form	Purpose	Examples
DDL	Data Definition Language	Defines structure	CREATE, ALTER, DROP
DML	Data Manipulation Language	Manipulates data	INSERT, UPDATE, DELETE
DCL	Data Control Language	Controls access	GRANT, REVOKE
TCL/DTL	Transaction Control Language	Manages transactions	COMMIT, ROLLBACK, SAVEPOINT

📘 Short Note (2–3 Marks)

SQL is a standard language used to interact with relational databases. SQL commands are grouped into DDL, DML, DCL, and TCL. DDL defines database structure, DML manipulates data, DCL controls permissions, and TCL manages transactions.

If you want, I can also provide:

- 📌 **SQL queries practice questions**
- 📌 **MCQs with answers**
- 📌 **Complete SQL notes as PDF**

DDL (CREATE/ALTER/DROP/TRUNCATE)

Here is a **clear, exam-oriented explanation of DDL statements (CREATE, ALTER, DROP, TRUNCATE)** with syntax + examples.

✅ DDL (Data Definition Language)

DDL commands are used to **define, modify, or delete database objects** such as tables, schemas, indexes, views, etc.
DDL statements **auto-commit** (changes cannot be rolled back).

The main DDL commands are:

1. **CREATE**
2. **ALTER**
3. **DROP**
4. **TRUNCATE**

1️⃣ CREATE

Used to **create** a new database object (table, database, view, index, etc.)

Syntax

sql Copy code

```sql
CREATE TABLE table_name (
    column_name datatype constraints,
    column_name datatype constraints
);
```

Example

sql Copy code

```sql
CREATE TABLE students (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    age INT,
    city VARCHAR(30)
);
```

2️⃣ ALTER

Used to **modify** the structure of an existing table.

Common ALTER Operations

✔ Add new column

sql Copy code

```sql
ALTER TABLE students
ADD email VARCHAR(100);
```

✔ Modify datatype

sql Copy code

```sql
ALTER TABLE students
MODIFY age BIGINT;
```

✔ Rename a column

sql Copy code

```sql
ALTER TABLE students
RENAME COLUMN city TO location;
```

✔ Drop a column

sql Copy code

```sql
ALTER TABLE students
DROP COLUMN email;
```

✔ Rename table

sql Copy code

```sql
ALTER TABLE students
RENAME TO student_details;
```

3️⃣ DROP

Used to **permanently delete** a table or database **along with all data**.

❗ Cannot be rolled back.

Syntax

sql Copy code

```sql
DROP TABLE table_name;
```

Example

sql

```sql
DROP TABLE student_details;
```

4️⃣ TRUNCATE

Used to **remove all records from a table**, but **keeps the table structure**.

❗ **Faster than DELETE (no row-by-row logging)**

❗ **Cannot be rolled back (auto-commit)**

Syntax

sql Copy code

```sql
TRUNCATE TABLE students;
```

📌 DDL vs DML (Quick Difference)

Feature	DDL	DML
Purpose	Table structure	Table data
Auto-commit	Yes	No
Rollback	❌ No	✔ YES
Examples	CREATE, ALTER, DROP	INSERT, UPDATE, DELETE

If you want, I can also explain **DML, DCL, TCL**, or generate **practice questions** for this topic.

Normalization, Need for Normalization

Below is a **clear, simple, exam-friendly explanation** of **Normalization** and the **Need for Normalization**, with examples.

⭐ Normalization

Normalization is a database design process used to organize data efficiently by reducing **redundancy** (duplicate data) and avoiding **anomalies** (errors during insert, update, delete).

Normalization breaks a large table into **smaller, related tables** and defines relationships between them.

It was introduced by **E. F. Codd** (the father of RDBMS).

⭐ Why Do We Need Normalization?

Normalization is required to solve the following problems:

1️⃣ To eliminate data redundancy (duplicate data)

If the same data is stored many times, it wastes space and causes inconsistency.

❌ Example (without normalization)

A single table:

StudentID	Name	Course	Course_Fee
1	Rahul	Java	5000
2	Priya	Python	6000
3	Rohan	Java	5000

→ "Java - 5000" is repeated multiple times.

→ If the fee changes, you must update all rows → **update anomaly**.

2️⃣ To avoid anomalies

Normalization prevents:

✔ Update Anomaly

One change requires multiple updates → risk of inconsistency.

✔ Insert Anomaly

You cannot insert some data unless other data is present.

Example:

Cannot add a new course unless a student enrolls in it.

✔ Delete Anomaly

Deleting a row removes important information.

Example:

If last student of Java is deleted → entire Java course info is lost.

3️⃣ To ensure data integrity

Data remains:

- ✔ Accurate
- ✔ Consistent
- ✔ Reliable
- ✔ Structured

4️⃣ To organize data efficiently

Normalized tables are:

- Smaller

- Faster to search
- Easier to maintain
- Less prone to errors

5️⃣ To improve query performance (in many cases)

Small tables → faster joins, faster lookups.

⭐ Normalization Process (Normal Forms)

Normalization is achieved through **Normal Forms**:

1. **1NF – First Normal Form**
2. **2NF – Second Normal Form**
3. **3NF – Third Normal Form**
4. **BCNF – Boyce-Codd Normal Form**
5. 4NF / 5NF (used in advanced cases)

If you want, I can explain each Normal Form with **examples + diagrams**.

⭐ Example: Normalization in Action

❌ Unnormalized Table

ID	Name	Courses
1	Amit	Java, Python

The "Courses" field has multiple values → violates 1NF.

✔️ After Normalization

Table: Student

StudentID	Name
1	Amit

Table: Courses

CourseID	CourseName
101	Java
102	Python

Table: StudentCourse (Many-to-Many)

StudentID	CourseID
1	101
1	102

Now the data is:

- ✔ clean
- ✔ organized
- ✔ no duplication
- ✔ no anomalies

If you want, I can explain:

- 📌 **1NF, 2NF, 3NF, BCNF with examples**
- 📌 **Differences between all Normal Forms**
- 📌 **Practice questions for Normalization (theory + diagrams)**

Normal Forms (1st NF, 2nd NF, 3rd NF, BCNF) with examples, Introduction to 4th and 5th NF

Here is a **clear, exam-focused, easy-to-understand** explanation of **1NF, 2NF, 3NF, BCNF**, plus **intro to 4NF & 5NF**, all with examples.

⭐ 1. First Normal Form (1NF)

A table is in **1NF** if:

- ✔ All values are **atomic** (no multiple values in one cell)
- ✔ No repeating groups / arrays
- ✔ Each record is unique

❌ Example (NOT in 1NF)

StudentID	Name	Courses
1	Amit	Java, Python
2	Riya	C, C++

Problem → **Courses has multiple values**.

✔ Convert to 1NF

StudentID	Name	Course
1	Amit	Java
1	Amit	Python
2	Riya	C
2	Riya	C++

Now, all values are atomic → **1NF achieved**.

⭐ 2. Second Normal Form (2NF)

A table is in **2NF** if:

✔ It is already in **1NF**
✔ **No partial dependency** exists
(Non-key attribute depends on **whole composite key**, not part of it)

❌ Example (NOT in 2NF)

StudentCourse Table

StudentID	CourseID	StudentName
1	101	Amit
1	102	Amit

Primary Key = (StudentID + CourseID)
But **StudentName depends only on StudentID**, not on CourseID → **Partial dependency**.

✔ Convert to 2NF

1. Student Table

StudentID	StudentName
1	Amit

2. StudentCourse Table

StudentID	CourseID
1	101
1	102

Now no non-key attribute depends on part of a composite key → **2NF achieved**.

⭐ 3. Third Normal Form (3NF)

A table is in **3NF** if:

✔ It is already in **2NF**

✔ No **transitive dependency**

(A non-key attribute should not depend on another non-key attribute)

❌ Example (NOT in 3NF)

EmpID	EmpName	DeptID	DeptName
1	Rahul	10	Sales
2	Priya	20	HR

Primary Key → EmpID

But **DeptName depends on DeptID**, which depends on EmpID:

EmpID → DeptID → DeptName

(This is transitive dependency)

✔ Convert to 3NF

1. Employee Table

EmpID	EmpName	DeptID
1	Rahul	10
2	Priya	20

2. Department Table

DeptID	DeptName
10	Sales
20	HR

Now → No transitive dependency → **3NF achieved**.

⭐ 4. Boyce-Codd Normal Form (BCNF)

BCNF is a stronger version of 3NF.

A table is in **BCNF** if:

✔ For every functional dependency **A → B**,
✔ **A must be a superkey**

Or simpler:

👉 **No non-trivial functional dependency should violate primary key constraints.**

❌ Example (NOT in BCNF)

Course	Teacher	Time
DBMS	Amit	10 AM
DBMS	Riya	11 AM

Functional dependencies:

1. Course → Teacher ❌ (Course is NOT a key)
2. Teacher → Time ✔
3. (Course, Time) → Teacher (Candidate key)

But **Course → Teacher** violates BCNF because Course is not a key.

✔ Convert to BCNF

1. CourseTeacher Table

Course	Teacher
DBMS	Amit

Course	Teacher
DBMS	Riya

2. TeacherTime Table

Teacher	Time
Amit	10 AM
Riya	11 AM

Now every determinant is a candidate key → **BCNF achieved**.

⭐ Introduction to 4th Normal Form (4NF)

4NF deals with **Multi-Valued Dependencies (MVD)**.

A table is in **4NF** if:

- ✔ It is already in **BCNF**
- ✔ It has **no multi-valued dependencies**

❌ Example (NOT in 4NF)

Student	Hobby	Language
Amit	Cricket	English
Amit	Cricket	Hindi
Amit	Music	English
Amit	Music	Hindi

MVD:

A student can have multiple hobbies,

and also multiple languages, independently.

✔ Convert to 4NF

1. StudentHobby

Student	Hobby
Amit	Cricket
Amit	Music

2. StudentLanguage

Student	Language
Amit	English
Amit	Hindi

Now → No independent multi-valued dependencies → **4NF achieved**.

⭐ Introduction to 5th Normal Form (5NF)

5NF (Project-Join Normal Form) deals with **join dependencies**.

A table is in **5NF** if:

- ✔ It is already in **4NF**
- ✔ It **cannot** be decomposed further without losing information
- ✔ It removes **complex join dependencies**

❌ Example (NOT in 5NF)

A complicated many-to-many relationship:

Supplier	Product	Project
S1	P1	J1
S1	P2	J1

Supplier	Product	Project
S1	P1	J2

This table may be broken into 3 tables **only if decomposition loses no data**.

✔ In 5NF (each represent separate relationship)

- Supplier–Product
- Supplier–Project
- Product–Project

If the original table can be rebuilt by joining these tables without losing data → **5NF achieved**.

⭐ Summary Table

Normal Form	Removes	Example Problem
1NF	Repeating groups	Multivalued fields
2NF	Partial dependency	Composite key issue
3NF	Transitive dependency	A → B → C
BCNF	Key dependency issues	Non-key determining key
4NF	Multi-valued dependency	Independent multivalues
5NF	Join dependency	Complex multi-table joins

If you want, I can also provide:

📌 **Practice questions**
📌 **ERD for normalization examples**
📌 **Diagram-based normalization problems**

DML(INSERT/UPDATE/DELETE)

Here is a **simple, clear, exam-oriented explanation** of **DML (Data Manipulation Language)** with **INSERT, UPDATE, DELETE** syntax + examples.

⭐ DML (Data Manipulation Language)

DML commands are used to **manipulate data stored inside tables**.
Unlike DDL, they **do NOT auto-commit** → you can **ROLLBACK** changes.

The main DML commands are:

✔ **INSERT**

✔ **UPDATE**

✔ **DELETE**

1️⃣ INSERT

Used to **add new rows** into a table.

✔ Syntax 1: Insert all columns

sql Copy code

```sql
INSERT INTO table_name
VALUES (value1, value2, value3, ...);
```

Example:

sql Copy code

```sql
INSERT INTO students
VALUES (1, 'Amit', 20, 'Pune');
```

✔ Syntax 2: Insert with column names

(Recommended)

sql Copy code

```sql
INSERT INTO table_name (col1, col2, col3)
VALUES (val1, val2, val3);
```

Example:

sql Copy code

```sql
INSERT INTO students (id, name, age)
VALUES (2, 'Priya', 22);
```

✔ Insert multiple rows

sql Copy code

```sql
INSERT INTO students (id, name, age)
VALUES
(3, 'Rohan', 21),
(4, 'Sneha', 23);
```

✔ Insert data from another table

sql Copy code

```sql
INSERT INTO new_table (id, name)
SELECT id, name FROM students;
```

2️⃣ UPDATE

Used to **modify existing records**.

✔️ Syntax

sql Copy code

```sql
UPDATE table_name
SET column1 = value1, column2 = value2
WHERE condition;
```

Example:

sql Copy code

```sql
UPDATE students
SET age = 25
WHERE id = 1;
```

✔️ Only student with ID = 1 gets updated.

❗ Without WHERE clause → updates all rows

sql Copy code

```sql
UPDATE students
SET age = 30;
```

(This is risky and usually avoided.)

3️⃣ DELETE

Used to **remove rows** from a table.

✔ Syntax

sql Copy code

```
DELETE FROM table_name
WHERE condition;
```

Example:

sql Copy code

```
DELETE FROM students
WHERE id = 2;
```

✔ Deletes the row where ID = 2.

❗ Without WHERE → deletes all rows

sql Copy code

```
DELETE FROM students;
```

⚠ Table structure remains → only data is removed.

⭐ DML vs DDL Quick Comparison

Feature	DML	DDL
Purpose	Manipulate **data**	Modify **structure**
Auto-commit	❌ No	✔ Yes
Rollback possible	✔ Yes	❌ No
Commands	INSERT, UPDATE, DELETE	CREATE, ALTER, DROP, TRUNCATE

If you want, I can also explain:

📌 **SELECT (DQL)**
📌 **DCL (GRANT/REVOKE)**
📌 **TCL (COMMIT/ROLLBACK/SAVEPOINT)**
📌 **Practice questions for DML**

MySQLDataTypes, Database Constraints (Primary Key, Unique, Not Null, Foreign Key, Default, Check*)

Here is a **clear, exam-focused explanation** of **MySQL Data Types** and **Database Constraints** with examples.

⭐ MySQL Data Types

MySQL data types are grouped into 3 major categories:

1️⃣ Numeric Data Types

Type	Description	Example
INT	Integer numbers	10, 500
BIGINT	Very large integers	9000000000
SMALLINT	Small ranges	200
TINYINT	Very small	1, -1
DECIMAL(p,s)	Exact decimal values	150.75

Type	Description	Example
FLOAT	Approx decimal values	10.25
DOUBLE	Larger floating point	1450.5673

✔ Use **DECIMAL** for *money*, to avoid rounding errors.

---

2 String / Character Data Types

Type	Description
CHAR(n)	Fixed-length string (faster)
VARCHAR(n)	Variable-length string
TEXT	Large text (up to 65k chars)
MEDIUMTEXT	Bigger text
LONGTEXT	Very large text
ENUM	One value from a set (e.g., 'Male','Female')
SET	Multiple values from predefined list

Examples:

sql Copy code

```sql
name VARCHAR(50)
gender ENUM('Male','Female','Other')
description TEXT
```

---

3 Date & Time Data Types

Type	Description	Example
DATE	YYYY-MM-DD	2025-11-17
TIME	HH:MM:SS	14:30:00
DATETIME	Date + time	2025-11-17 14:30:00
TIMESTAMP	Auto-store current time	CURRENT_TIMESTAMP
YEAR	Year only	2025

⭐ Database Constraints

Constraints ensure **data integrity** and **valid data input**.

The main MySQL constraints:

- **PRIMARY KEY**
- **UNIQUE**
- **NOT NULL**
- **FOREIGN KEY**
- **DEFAULT**
- **CHECK** (MySQL 8+)

Let's explain each:

1️⃣ PRIMARY KEY (PK)

Uniquely identifies each row in a table.

✔ Must be **unique**
✔ Cannot be **null**
✔ Only **one primary key** per table
✔ Can be single or composite

Example:

sql Copy code

```sql
CREATE TABLE students (
  id INT PRIMARY KEY,
  name VARCHAR(50)
);
```

2️⃣ UNIQUE Constraint

Ensures **no duplicate values**, but allows **one NULL value**.

Example:

sql Copy code

```sql
CREATE TABLE users (
  email VARCHAR(100) UNIQUE
);
```

3️⃣ NOT NULL

Prevents NULL values → column must always contain data.

Example:

sql Copy code

```sql
name VARCHAR(50) NOT NULL
```

4️⃣ FOREIGN KEY

Creates a relationship between two tables.

Ensures **referential integrity**.

Example:

Department Table:

sql Copy code

```sql
CREATE TABLE department (
  dept_id INT PRIMARY KEY,
  dept_name VARCHAR(50)
);
```

Employee Table:

sql Copy code

```sql
CREATE TABLE employee (
  emp_id INT PRIMARY KEY,
  name VARCHAR(50),
  dept_id INT,
  FOREIGN KEY (dept_id) REFERENCES department(dept_id)
);
```

✔ Ensures you cannot insert an employee with a non-existing dept_id.
✔ Also prevents deleting a department if employees depend on it (unless using CASCADE).

5️⃣ DEFAULT Constraint

Assigns default value when no value is supplied.

Example:

sql Copy code

```sql
status VARCHAR(20) DEFAULT 'Active'
```

If user does not provide status → it becomes "Active".

6️⃣ CHECK Constraint *(MySQL 8.0+)*

Ensures that the column value meets a specific condition.

Example:

sql Copy code

```sql
age INT CHECK (age >= 18);
```

Another example:

sql Copy code

```sql
salary DECIMAL(10,2) CHECK (salary > 0);
```

⭐ Example Table with All Constraints

sql Copy code

```sql
CREATE TABLE students (
  id INT PRIMARY KEY,
  name VARCHAR(50) NOT NULL,
  age INT CHECK (age >= 18),
  email VARCHAR(100) UNIQUE,
  status VARCHAR(20) DEFAULT 'Active',
  dept_id INT,
  FOREIGN KEY (dept_id) REFERENCES department(dept_id)
);
```

⭐ If you want, I can also explain:

📌 DCL (GRANT/REVOKE)
📌 TCL (COMMIT/ROLLBACK/SAVEPOINT)
📌 SELECT Query (DQL)
📌 Views, Indexes, Joins
📌 Exercises + MCQs for Data Types & Constraints

Aggregate Functions, Grouping Things Together (Group By, Having)

Here is a **simple, exam-friendly explanation** of **Aggregate Functions**, **GROUP BY**, and **HAVING** with examples.

⭐ Aggregate Functions in SQL

Aggregate functions perform **calculations on multiple rows** and return **a single value**.

Common aggregate functions:

Function	Meaning
COUNT()	Counts rows
SUM()	Adds values
AVG()	Calculates average
MIN()	Lowest value
MAX()	Highest value

These are mostly used with **GROUP BY**.

⭐ 1️⃣ Aggregate Function Examples

Assume a table **sales**:

id	product	amount
1	Mobile	10000
2	Laptop	40000

id	product	amount
3	Mobile	15000
4	Laptop	30000

✔ COUNT()

sql Copy code

```sql
SELECT COUNT(*)
FROM sales;
```

Returns total number of rows.

✔ SUM()

sql Copy code

```sql
SELECT SUM(amount)
FROM sales;
```

✔ AVG()

sql Copy code

```sql
SELECT AVG(amount) FROM sales;
```

✔ MIN() / MAX()

sql Copy code

```sql
SELECT MIN(amount), MAX(amount)
```

```
FROM sales;
```

⭐ Grouping Things Together — GROUP BY

GROUP BY is used to **group rows based on one or more columns**, and then apply aggregate functions.

✔ Example: Total sales per product

sql Copy code

```
SELECT product, SUM(amount)
FROM sales
GROUP BY product;
```

Output:

product	SUM(amount)
Mobile	25000
Laptop	70000

✔ Example: Count of each product

sql Copy code

```
SELECT product, COUNT(*)
FROM sales
GROUP BY product;
```

✔ Grouping by multiple columns

sql Copy code

```sql
SELECT product, amount, COUNT(*)
FROM sales
GROUP BY product, amount;
```

⭐ HAVING Clause

HAVING is used to **apply conditions on groups** (after GROUP BY).

📌 **WHERE → filters rows before grouping**
📌 **HAVING → filters groups after grouping**

❌ Invalid (cannot use aggregate in WHERE):

sql Copy code

```sql
SELECT product, SUM(amount)
FROM sales
WHERE SUM(amount) > 30000;  -- ❌ Error
```

✔️ Correct: Use HAVING

sql Copy code

```sql
SELECT product, SUM(amount)
FROM sales
GROUP BY product
HAVING SUM(amount) > 30000;
```

Output:

product	SUM(amount)
Laptop	70000

⭐ Another Example (common exam question)

🔹 Find products sold more than once:

sql Copy code

```sql
SELECT product, COUNT(*)
FROM sales
GROUP BY product
HAVING COUNT(*) > 1;
```

Output:

product	COUNT(*)
Mobile	2
Laptop	2

⭐ WHERE vs HAVING (Very Important)

Feature	WHERE	HAVING
Works on	Rows	Groups
Used with	Any select query	Always with GROUP BY
Aggregate functions	❌ Not allowed	✔ Allowed
Executes	Before grouping	After grouping

⭐ Summary

Aggregate Functions:

- COUNT()
- SUM()
- AVG()

- MIN()
- MAX()

GROUP BY:

- Groups rows
- Used with aggregates

HAVING:

- Applies conditions on grouped data
- Works with aggregates (e.g., SUM, COUNT)

If you want, I can also explain:

📌 Joins (Inner/Left/Right/Full)
📌 Subqueries
📌 Views, Indexes
📌 Stored Procedures & Functions
📌 Practice questions for GROUP BY + HAVING

LIKE Operator, DISTINCT, Sorting (Order by clause)

Here is a **clear, exam-focused explanation** of the **LIKE operator, DISTINCT keyword, and ORDER BY (sorting)** with syntax + examples.

⭐ 1. LIKE Operator

The **LIKE** operator is used for **pattern matching** in SQL.
It works with **wildcards**:

✔ Wildcards in LIKE:

Symbol	Meaning
%	Matches **0 or more characters**
_	Matches **exactly 1 character**

✔️ Syntax

sql Copy code

```sql
SELECT columns
FROM table
WHERE column LIKE pattern;
```

⭐ Examples

1️⃣ Find names starting with 'A'

sql Copy code

```sql
SELECT name FROM students
WHERE name LIKE 'A%';
```

Matches → "Amit", "Anita", "Ayesha"

2️⃣ Find names ending with 'a'

sql Copy code

```sql
SELECT name FROM students
WHERE name LIKE '%a';
```

Matches → "Riya", "Sneha"

3️⃣ Find names containing 'it'

sql Copy code

```sql
SELECT name FROM students
WHERE name LIKE '%it%';
```

Matches → "Amit", "Nikita"

4️⃣ Find names where 2nd letter is 'a'

sql Copy code

```sql
SELECT name FROM students
WHERE name LIKE '_a%';
```

Matches → "Ravi", "Rahul", "Manish" (second letter is 'a')

5️⃣ Find 3-letter names

sql Copy code

```sql
SELECT name FROM students
WHERE name LIKE '___';
```

⭐ 2. DISTINCT Keyword

DISTINCT is used to **remove duplicate values** from the result.

✔️ Syntax

sql Copy code

```sql
SELECT DISTINCT column_name
FROM table;
```

⭐ Examples

1️⃣ Get unique cities

sql Copy code

```sql
SELECT DISTINCT city
FROM students;
```

2️⃣ Get unique combinations of city & course

sql Copy code

```sql
SELECT DISTINCT city, course
FROM students;
```

⭐ 3. ORDER BY (Sorting)

Used to **sort the result set**.

✔ Ascending (default)

sql Copy code

```sql
ORDER BY column ASC;
```

✔ Descending

sql Copy code

```sql
ORDER BY column DESC;
```

You can sort by multiple columns.

⭐ Examples

1️⃣ Sort students by name (ascending)

sql Copy code

```sql
SELECT * FROM students
ORDER BY name;
```

(Same as `ORDER BY name ASC`)

2️⃣ Sort by age descending

sql Copy code

```sql
SELECT * FROM students
ORDER BY age DESC;
```

3️⃣ Sort by city first, then by name

sql Copy code

```sql
SELECT * FROM students
ORDER BY city ASC, name ASC;
```

4️⃣ Sort by marks highest first

sql Copy code

```sql
SELECT name, marks
FROM results
ORDER BY marks DESC;
```

⭐ Combined Example (Important)

Get unique product names starting with 'A', sorted by price:

sql Copy code

```sql
SELECT DISTINCT product_name, price
FROM products
WHERE product_name LIKE 'A%'
ORDER BY price DESC;
```

⭐ Summary Table

Feature	Purpose	Example
LIKE	Pattern search	`name LIKE 'A%'`
DISTINCT	Remove duplicates	`SELECT DISTINCT city`
ORDER BY	Sorting results	`ORDER BY age DESC`

If you want, I can explain:

📌 LIMIT clause
📌 Joins
📌 Subqueries
📌 Views & Indexes
📌 Practice questions for LIKE, DISTINCT, ORDER BY

Relational Algebra Operations (Selection, Projection, Union, Intersect*, Minus*, Cross/Cartesian)

Here is a clear, exam-oriented explanation of **Relational Algebra Operations** with definitions + symbols + examples.

Relational Algebra Operations

Relational Algebra is a **theoretical query language** used to operate on relations (tables). SQL is based on relational algebra.

---

1. Selection (σ)

Purpose: Selects **rows** based on a condition (filters tuples).

Symbol: σ (sigma)

Syntax:

σ *condition* (Relation)

Example:

Employee(**ID**, Name, Salary)

Query: Get employees whose salary > 50000
σ Salary > 50000 (Employee)

---

2. Projection (π)

Purpose: Selects **columns** from a table.

Symbol: π (pi)

Syntax:

π *column_list* (Relation)

Example:

Employee(**ID**, Name, Salary)

Query: List only names
π Name (Employee)

---

3. Union (∪)

Purpose: Combines the tuples of two relations, **removes duplicates**.

Conditions (very important):

Two relations must be:

✔ Same number of columns

✔ Corresponding columns must be of same type

Example:

A = Students from Dept 1

B = Students from Dept 2

A ∪ B → All unique students

---

4. Intersection (∩) *(Sometimes represented as ∩ or INTERSECT)*

Purpose: Returns **common tuples** in both relations.

Example:

A = Students playing Cricket

B = Students playing Football

A ∩ B → Students who play **both** Cricket and Football

---

5. Minus / Difference (−)

Purpose: Returns tuples in A **but not in B**.

Example:

A = All Students

B = Students who passed

A − B → Students who FAILED

---

6. Cartesian Product (×)

Purpose: Combines every row of first table with every row of second table.

Example:

A has 3 rows

B has 4 rows

A × B → 3 × 4 = **12 rows**

Usefulness:

Cartesian product is the basis for **joins**.

Summary Table

Operation	Symbol	Works On	Purpose
Selection	σ	Rows	Filter rows
Projection	π	Columns	Select specific columns
Union	∪	Rows	Combine and remove duplicates
Intersection	∩	Rows	Common rows
Minus	−	Rows	Rows in A not in B
Cartesian Product	×	Rows	Combine each row of A with each row of B

If you want, I can also explain:

- ✔ **Joins in Relational Algebra**
- ✔ **Examples with diagrams**
- ✔ **Practice questions**

Would you like those?

Joins (Eqvi, Inner, Outer, Natural, Cross), SQL Standard Syntax for Joins

Below is a clear, exam-oriented explanation of **Joins in DBMS**, including **types of joins**, **examples**, and **SQL standard syntax**.

JOINS in SQL & Relational Algebra

A **JOIN** combines rows from two or more tables based on a related column (usually a **foreign key**).

Assume two sample tables:

Students

sid	name	dept
1	Aditi	CS
2	Raj	IT
3	Meera	CS
4	Mohan	ME

Departments

dept	hod
CS	Sharma
IT	Verma
EE	Rakesh

1. Equi Join

A join that uses an **equality (=) comparison** between columns.

SQL Syntax

sql Copy code

```sql
SELECT *
FROM Students S
JOIN Departments D
ON S.dept = D.dept;
```

This is basically an **inner join using = condition**.

---

2. Inner Join

Returns **only matching rows** from both tables.

SQL Syntax

sql Copy code

```sql
SELECT *
FROM Students S
INNER JOIN Departments D
ON S.dept = D.dept;
```

Result: Only students whose dept exists in Departments table.

---

3. Outer Joins

Outer joins return **matching rows + non-matching rows** based on the join type.

There are **3 types**:

3.1 LEFT OUTER JOIN

Returns **all rows from left table**, plus matching rows from right table.

sql Copy code

```sql
SELECT *
FROM Students S
LEFT JOIN Departments D
ON S.dept = D.dept;
```

Students with departments not present in the Departments table will have **NULL** in right table columns.

3.2 RIGHT OUTER JOIN

Returns **all rows from right table**, plus matching rows from left table.

sql Copy code

```sql
SELECT *
FROM Students S
RIGHT JOIN Departments D
ON S.dept = D.dept;
```

Departments with no students will appear with NULLs.

3.3 FULL OUTER JOIN

Returns:

- ✔ Matching rows
- ✔ Unmatched rows from left
- ✔ Unmatched rows from right

sql Copy code

```sql
SELECT *
FROM Students S
```

```
FULL OUTER JOIN Departments D
ON S.dept = D.dept;
```

⚠️ Note: **MySQL does not support FULL JOIN directly** (use UNION workaround).

---

4. Natural Join

A join that automatically uses **columns with the same name** in both tables.

No need to specify ON condition.

SQL Syntax

sql Copy code

```
SELECT *
FROM Students
NATURAL JOIN Departments;
```

Important:

- If column names differ (example: dept_code vs dept), natural join will not work.
- It removes duplicate columns in result.

---

5. Cross Join (Cartesian Product)

Combines **every row** of table A with **every row** of table B.

SQL Syntax

sql Copy code

```
SELECT *
FROM Students
CROSS JOIN Departments;
```

If Students has 4 rows and Departments has 3 rows → result = 12 rows.

---

6. SQL Standard JOIN Syntax (Summary)

INNER JOIN

sql Copy code

```sql
SELECT ...
FROM A
INNER JOIN B ON A.col = B.col;
```

LEFT JOIN

sql Copy code

```sql
SELECT ...
FROM A
LEFT JOIN B ON A.col = B.col;
```

RIGHT JOIN

sql Copy code

```sql
SELECT ...
FROM A
RIGHT JOIN B ON A.col = B.col;
```

FULL JOIN

sql Copy code

```sql
SELECT ...
FROM A
FULL JOIN B ON A.col = B.col;
```

NATURAL JOIN

sql Copy code

```sql
SELECT ...
FROM A
```

```
NATURAL JOIN B;
```

CROSS JOIN

sql Copy code

```
SELECT ...
FROM A
CROSS JOIN B;
```

EQUIJOIN (same as default join with =)

sql Copy code

```
SELECT ...
FROM A
JOIN B ON A.col = B.col;
```

Diagram Summary

Join Type	What it Returns
Inner Join	Only matching rows
Left Join	All rows from left + matches
Right Join	All rows from right + matches
Full Join	All rows from both tables
Natural Join	Auto-matching columns by name
Cross Join	Cartesian product

If you want, I can also provide:

✔ Join diagrams (Venn diagrams)

✔ 20 practice MCQs on SQL joins
✔ Real-world examples with tables

Would you like those?

Copying table structure/data, Sequences (AUTO_INCREMENT)

Here is a clear and exam-friendly explanation of **Copying Table Structure/Data in MySQL** and **Sequences (AUTO_INCREMENT)** with examples.

✅ 1. Copying Table Structure (Without Data)

To create a new table with the **same structure** as an existing table:

Method 1: Using LIKE

sql Copy code

```sql
CREATE TABLE new_table LIKE old_table;
```

✔ Copies structure
✔ Copies indexes (PK, Unique), default values
❌ Does NOT copy data

✅ 2. Copying Table Data (Without Structure)

To copy only the **data** into an existing table with matching structure:

sql Copy code

```sql
INSERT INTO new_table
SELECT * FROM old_table;
```

⚠️ `new_table` must already exist.

✅ 3. Copying Both Structure + Data into a New Table

You can create a new table and copy data in one command.

Method 1: With WHERE clause

sql Copy code

```sql
CREATE TABLE new_table AS
SELECT * FROM old_table;
```

✔ Copies data
✔ Creates table automatically
❌ Does NOT copy constraints (PK, indexes)

Method 2: Copying partial columns

sql Copy code

```sql
CREATE TABLE new_table AS
SELECT id, name FROM old_table;
```

✅ 4. Copying Only Selected Records

sql Copy code

```sql
CREATE TABLE cs_students AS
SELECT * FROM students WHERE dept = 'CS';
```

This creates a new table containing only CS students.

🔁 5. Copying Table With Constraints – Best Method

If you want **exact replication**:

sql Copy code

```sql
CREATE TABLE new_table LIKE old_table;
INSERT INTO new_table SELECT * FROM old_table;
```

- ✔ Structure
- ✔ Data
- ✔ Constraints
- ✔ Indexes
- ✔ Auto_increment

🌟 AUTO_INCREMENT (Sequences in MySQL)

In other DBMS (Oracle, PostgreSQL), a **Sequence** is used to generate unique numbers.

MySQL uses **AUTO_INCREMENT**, which automatically generates increasing IDs.

✅ 1. Creating AUTO_INCREMENT column

sql Copy code

```sql
CREATE TABLE students (
    id INT AUTO_INCREMENT,
    name VARCHAR(100),
    PRIMARY KEY(id)
);
```

- ✔ Automatically generates 1,2,3...
- ✔ No need to insert ID manually

✅ 2. Inserting Without Giving ID

sql Copy code

```sql
INSERT INTO students(name) VALUES ('Aditi');
INSERT INTO students(name) VALUES ('Rahul');
```

Generated IDs:

id	name
1	Aditi
2	Rahul

✅ 3. Setting AUTO_INCREMENT Starting Value

Default starts at **1**.

To start from 500:

sql Copy code

```sql
ALTER TABLE students AUTO_INCREMENT = 500;
```

Next insert → id = 500.

✅ 4. Getting the Last AUTO_INCREMENT Value

For the current session:

sql Copy code

```sql
SELECT LAST_INSERT_ID();
```

✅ 5. Reset AUTO_INCREMENT After Deleting Rows

If table is empty:

sql Copy code

```sql
TRUNCATE TABLE students;
```

AUTO_INCREMENT resets to **1**.

If you only delete rows:

sql Copy code

```sql
DELETE FROM students;
ALTER TABLE students AUTO_INCREMENT = 1;
```

📌 Summary

Task	SQL Command
Copy structure only	`CREATE TABLE new LIKE old;`
Copy data only	`INSERT INTO new SELECT * FROM old;`
Copy structure + data	`CREATE TABLE new AS SELECT * FROM old;`
Copy exact replica	LIKE + INSERT
Create auto sequence	`AUTO_INCREMENT`
Reset sequence	`ALTER TABLE table AUTO_INCREMENT = n;`

If you want, I can also give:

- ✔ Practice questions
- ✔ Viva questions
- ✔ MCQs on this topic

Would you like those?

Subquery, Correlated Subquery, EXISTS/NOT EXISTS

Below is a **clear, exam-ready explanation** of **Subquery, Correlated Subquery, EXISTS & NOT EXISTS** with examples.

✅ 1. Subquery (Nested Query)

A **subquery** is a query inside another SQL query.

Used for:

- ✔ Filtering
- ✔ Comparisons
- ✔ Checking existence
- ✔ Returning values to outer query

Types of Subqueries

- **Single-row subquery** → returns 1 value
- **Multi-row subquery** → returns multiple values
- **Scalar subquery** → returns exactly 1 value
- **Nested subquery** → inside another subquery

📌 Example 1: Subquery (Basic)

Get students who have marks greater than the average marks.

sql Copy code

```sql
SELECT name
FROM students
WHERE marks > (
    SELECT AVG(marks)
    FROM students
);
```

- ✔ Subquery runs first
- ✔ Outer query uses result

📌 Example 2: Multi-row Subquery

Find employees who work in departments located in 'Pune'.

sql Copy code

```sql
SELECT name
FROM employee
WHERE dept_id IN (
    SELECT dept_id
    FROM department
    WHERE location = 'Pune'
);
```

🚀 2. Correlated Subquery

A **correlated subquery** depends on the **outer query**.

It runs **once for each row** of the outer query.

Key Point:

Subquery references columns from outer query.

📌 Example: Correlated Subquery

Find employees whose salary is **above the average salary** of their **own department**.

sql Copy code

```sql
SELECT e1.name, e1.salary
FROM employees e1
WHERE e1.salary > (
    SELECT AVG(e2.salary)
    FROM employees e2
    WHERE e1.dept = e2.dept
);
```

- ✔ e1 (outer query) row value is used inside subquery
- ✔ Subquery executes once per employee

💡 Difference: Subquery vs Correlated Subquery

Subquery	Correlated Subquery
Executes once	Executes for each row
Independent	Depends on outer query
Faster	Slower
Example: AVG of all students	AVG for each department

🟩 3. EXISTS & NOT EXISTS

`EXISTS` is a logical operator that returns **TRUE if subquery returns at least one row.**

Used for:

- ✔ Fast membership checking
- ✔ Correlated queries
- ✔ Optimization

🚀 EXISTS Example

Find customers who have placed at least one order.

sql Copy code

```sql
SELECT c.name
FROM customers c
WHERE EXISTS (
    SELECT 1
    FROM orders o
    WHERE c.cid = o.cid
);
```

- ✔ If the subquery finds even **1 matching row**, EXISTS = TRUE
- ✔ `SELECT 1` is used because value doesn't matter

🚫 NOT EXISTS Example

Find customers who have **never placed any order**.

sql Copy code

```
SELECT c.name
FROM customers c
WHERE NOT EXISTS (
    SELECT 1
    FROM orders o
    WHERE c.cid = o.cid
);
```

- ✔ Returns customers with **0 orders**

🔥 EXISTS vs IN

EXISTS	IN
Faster for large datasets	Slower for large list
Checks **row existence**	Compares values
Uses correlated subquery	Subquery independent

- ✔ EXISTS is better for checking relationships
- ✔ IN is used for comparing column values

📝 Summary Table

Concept	Meaning	Executes
Subquery	A query inside another query	Once

Concept	Meaning	Executes
Correlated Subquery	Uses values from outer query	Per row
EXISTS	True if subquery returns any row	Typically fastest
NOT EXISTS	True if subquery returns no rows	Uses anti-join logic

---

If you want, I can also provide:

- ✔ Practice questions (10 or 20)
- ✔ MCQs for exam
- ✔ Real-world examples

Just tell me!

TCL Commands (Commit/Rollback/Savepoint), DCL Commands (GRANT/REVOKE/GRANT OPTION)

Here is a **clean, exam-oriented explanation** of **TCL Commands (COMMIT, ROLLBACK, SAVEPOINT)** and **DCL Commands (GRANT, REVOKE, GRANT OPTION)** with examples.

✅ Transaction Control Language (TCL)

Used to **manage transactions** in a database.

A **transaction** = a group of SQL operations treated as a **single logical unit**.

🌟 1. COMMIT

Permanently saves all changes made during the transaction.

Syntax

sql Copy code

```sql
COMMIT;
```

Example

sql Copy code

```sql
INSERT INTO accounts VALUES (1, 'Aditi', 5000);
COMMIT;
```

After COMMIT → Data **permanently stored** (cannot be undone).

🌟 2. ROLLBACK

Undo changes since the last COMMIT or SAVEPOINT.

Syntax

sql Copy code

```sql
ROLLBACK;
```

Example

sql Copy code

```sql
DELETE FROM accounts WHERE id = 5;
ROLLBACK;
```

Effect → The delete operation is undone.

🌟 3. SAVEPOINT

Creates a **checkpoint** inside a transaction.

You can roll back to a specific SAVEPOINT instead of the whole transaction.

Syntax

sql Copy code

```sql
SAVEPOINT sp1;
```

Example (Full Flow)

sql Copy code

```sql
START TRANSACTION;

UPDATE accounts SET balance = balance - 1000 WHERE id = 1;
SAVEPOINT s1;

UPDATE accounts SET balance = balance + 1000 WHERE id = 2;
SAVEPOINT s2;

-- Rollback second operation only
ROLLBACK TO s1;

COMMIT;
```

- ✔ Operation after `s1` is undone
- ✔ Operation before `s1` is kept
- ✔ `COMMIT` finalizes remaining changes

📜 Summary of TCL Commands

Command	Meaning
COMMIT	Save changes permanently
ROLLBACK	Undo changes since COMMIT
SAVEPOINT	Set a checkpoint in a transaction

Command	Meaning
ROLLBACK TO sp	Undo till that savepoint

🛡️ Data Control Language (DCL)

Used for **controlling access** (permissions) in a DBMS.

🌟 1. GRANT

Gives permission to a user.

Syntax

sql Copy code

```sql
GRANT privilege_list ON table_name TO user;
```

Example

Give user `rahul` permission to SELECT and INSERT on `employees` :

sql Copy code

```sql
GRANT SELECT, INSERT ON employees TO 'rahul';
```

🌟 2. REVOKE

Removes previously granted permissions.

Syntax

sql Copy code

```sql
REVOKE privilege_list ON table_name FROM user;
```

Example

Remove INSERT permission from `rahul` :

sql Copy code

```sql
REVOKE INSERT ON employees FROM 'rahul';
```

🌟 3. GRANT OPTION

Allows the user to **grant permissions to others**.

Syntax

sql Copy code

```sql
GRANT SELECT ON employees TO 'rahul' WITH GRANT OPTION;
```

Meaning → `rahul` can now grant SELECT permission to someone else.

🗒️ Privileges commonly used in MySQL

- `SELECT`
- `INSERT`
- `UPDATE`
- `DELETE`
- `CREATE`
- `DROP`
- `ALTER`
- `INDEX`

📝 Summary of DCL Commands

Command	Function
GRANT	Assigns privileges to user
REVOKE	Removes privileges

Command	Function
GRANT OPTION	Allows user to pass privileges to others

If you want, I can also provide:

- ✔ A table of all privileges (DB/table/user-level)
- ✔ Practice MCQs
- ✔ Short 2–3 mark exam answers for DBMS theory

Just tell me!

Views, Types of Views, Simple and Complex Views

Below is a **clear, exam-oriented explanation** of **Views**, **Types of Views**, **Simple View**, and **Complex View** with examples.

🌟 1. What is a View?

A **View** is a **virtual table** created using a SQL query.
It does **not store data physically**—it displays data stored in underlying tables.

Syntax

sql Copy code

```sql
CREATE VIEW view_name AS
SELECT columns
FROM table
WHERE condition;
```

✔ Key Points

- Acts like a table but **not stored permanently** (only its definition is stored).
- Useful for **security**, **simplicity**, and **hiding complexity**.
- When you query a view, it executes its **SELECT** statement internally.

🌟 2. Types of Views

Type	Meaning
1. Simple View	Created from **one table**, no joins, no group functions
2. Complex View	Created from **multiple tables**, joins, group functions
*3. Materialized View (Oracle)**	Physically stores data (Not in MySQL)
4. Inline View	A subquery used in FROM clause (temporary view)
5. Updatable View	Allows INSERT/UPDATE/DELETE
6. Read-only View	Cannot modify data

(*Materialized view is not supported directly in MySQL, but included for theory exams.)

🌟 3. Simple View

A **simple view** is created from **only one base table** and usually includes:

✔ No group functions
✔ No DISTINCT
✔ No joins
✔ Not always read-only — can often be updated

Example

sql Copy code

```sql
CREATE VIEW cs_students AS
SELECT id, name, dept
```

```
FROM students
WHERE dept = 'CS';
```

Features of Simple View

- Based on **single table**
- Can be **updatable** (if no restrictions)
- Used to hide columns or show filtered data

🌟 4. Complex View

A **complex view** is created from:

- ✔ Multiple tables
- ✔ Joins
- ✔ Aggregate functions (SUM, AVG, COUNT)
- ✔ GROUP BY
- ✔ DISTINCT, expressions, subqueries

This type of view is usually **read-only**.

Example 1: Join-based View

sql Copy code

```
CREATE VIEW student_details AS
SELECT s.name, d.hod
FROM students s
JOIN departments d
ON s.dept = d.dept;
```

Example 2: Group Function View

sql Copy code

```
CREATE VIEW dept_average AS
SELECT dept, AVG(marks) AS avg_marks
FROM students
GROUP BY dept;
```

Features of Complex View

- May involve **multiple tables**
- Often **not updatable**
- Used for reporting and summarization
- Provides abstraction for complicated queries

🌟 5. Updatable vs Non-updatable Views

✔️ Updatable View

A view can be updated when:

- Based on **one table**
- No aggregate functions
- No DISTINCT
- No GROUP BY
- No UNION
- Primary key remains visible

✗ Non-Updatable View

A view becomes read-only when:

- Uses **joins**
- Has **aggregations**
- Uses **DISTINCT**, **GROUP BY**, **UNION**
- Uses derived columns

🌟 6. Modifying a View

ALTER VIEW

sql Copy code

```sql
ALTER VIEW cs_students AS
SELECT id, name
```

```
FROM students
WHERE dept = 'CS';
```

🌟 7. Dropping a View

sql Copy code

```
DROP VIEW cs_students;
```

📝 Summary Table

Feature	Simple View	Complex View
No. of base tables	1	1 or more
Contains aggregates	No	Yes (possible)
Contains joins	No	Yes
Updatable	Mostly yes	Mostly no
Use case	Hide data, filter rows	Reports, calculations, joins

If you want, I can also provide:

- ✔ Diagram of simple vs complex view
- ✔ 10 practical questions on views
- ✔ Viva questions with answers

Just tell me!

Indexes, Benefit of Indexes, Type of Indexes, Temporary Tables

Below is a **clean, exam-ready explanation** of **Indexes**, **Benefits of Indexes**, **Types of Indexes**, and **Temporary Tables** with examples (MySQL-focused but applicable to DBMS theory).

📌 1. Indexes

An **Index** is a data structure (usually a **B-Tree**) used to **speed up data retrieval** from a database table.

Indexes work like the **index in a book**:

- Without index → you search page by page
- With index → you directly go to the page

✔ Index improves:

- SELECT performance
- WHERE clause
- JOIN operations
- ORDER BY and GROUP BY operations

✗ Index slows down:

- INSERT
- UPDATE
- DELETE

  Because index must also be updated.

📌 2. Benefits of Indexes

Benefit	Explanation
Faster search	Speeds up SELECT queries
Faster sorting	ORDER BY uses indexes
Faster join	Joins through indexed keys are faster
Enforces uniqueness	UNIQUE and PRIMARY KEY auto-create indexes
Reduced I/O	Fewer disk reads needed

📌 3. Types of Indexes

In MySQL, important index types are:

---

1. Primary Key Index

Automatically created when a **PRIMARY KEY** is defined.

Example

sql Copy code

```sql
CREATE TABLE student (
    id INT PRIMARY KEY,
    name VARCHAR(50)
);
```

Creates a unique, clustered index.

---

2. Unique Index

Ensures all values in the column are **unique**.

sql Copy code

```sql
CREATE UNIQUE INDEX idx_email ON users(email);
```

- ✔ Prevents duplicates
- ✔ Allows NULL (unless defined otherwise)

---

3. Normal (Non-Unique) Index

Used to speed up queries on frequently searched columns.

sql Copy code

```sql
CREATE INDEX idx_name ON student(name);
```

---

4. Composite Index (Multi-column index)

Index on **multiple columns**.

sql Copy code

```sql
CREATE INDEX idx_name_dept ON student(name, dept);
```

Used when queries search by both columns.

---

5. Full-text Index

Used for searching large text fields (blogs, articles, messages).

sql Copy code

```sql
CREATE FULLTEXT INDEX ft_content ON posts(content);
```

Supports advanced text search: MATCH() AGAINST()

---

6. Spatial Index

Used for geographic data types (GIS, maps).

sql Copy code

```sql
CREATE SPATIAL INDEX idx_location ON places(geo_point);
```

---

7. Clustered vs Non-Clustered Index (DBMS Theory)

Type	Meaning
Clustered Index	Data is physically ordered based on index (1 per table)
Non-Clustered Index	Separate structure storing pointers to table rows

MySQL InnoDB:

✔ Primary Key → Clustered Index

✔ Others → Non-clustered

📌 4. Temporary Tables

A **Temporary Table** is a table that exists **only during the session** or until you close the connection.

✔ Used for:

- Storing intermediate results
- Large calculations
- Complex joins
- Session-specific data

🔹 Creating a Temporary Table

sql Copy code

```sql
CREATE TEMPORARY TABLE temp_students (
    id INT,
    name VARCHAR(50)
);
```

🔹 Inserting Data

sql Copy code

```sql
INSERT INTO temp_students VALUES (1, 'Aditi');
```

🔹 Using SELECT INTO (Copy data into temporary table)

sql Copy code

```sql
CREATE TEMPORARY TABLE cs_students AS
SELECT * FROM students WHERE dept = 'CS';
```

🔹 Temporary Table Auto-Delete

- Automatically removed when:
  - ✔️ Session ends
  - ✔️ Connection closes
- You can also drop manually:

sql Copy code

```sql
DROP TEMPORARY TABLE temp_students;
```

📌 Advantages of Temporary Tables

Advantage	Explanation
Session-specific	No conflict with other users
Faster processing	Used for intermediate steps
Auto-deletion	Saves memory

Advantage	Explanation
Can index	Can add indexes inside temp tables

📌 Difference: Index vs Temporary Table

Index	Temporary Table
Speeds read performance	Stores temporary data
Permanent	Exists temporarily
Auto-created for PK/Unique	Created manually
Uses B-Tree/Hash	Works like normal table

📘 Quick Revision Summary

✔ Indexes

- Improve performance
- Types: Primary, Unique, Normal, Composite, Full-text, Spatial
- Primary Key → Clustered index
- Too many indexes slow down write operations

✔ Temporary Tables

- Exist only for session
- Useful for intermediate operations

If you want, I can also provide:

✔ 10 MCQs on Indexes & Temporary Tables
✔ Short notes format (perfect for notebook)
✔ Diagram-based explanation

Just tell me!

ACID Properties, Concept of Database Instance and Schema

Here is a clear, simple, exam-oriented explanation of **ACID Properties** and **Database Instance vs Schema**:

✅ ACID Properties

ACID refers to four key properties that ensure **reliable and consistent transaction processing** in a database.

1️⃣ Atomicity

- A transaction is **all or nothing**.
- Either every step is completed, or none are.
- Prevents partial updates.

Example:

Money transfer from A → B

- Deduct ₹500 from A
- Add ₹500 to B

  If adding fails, deduction must also rollback.

2️⃣ Consistency

- Transaction takes the database **from one valid state to another valid state**.
- Ensures all rules, constraints, triggers are followed.

Example:

A constraint says: *Age cannot be negative*.

If a transaction tries to insert age = –5 → rejected for violating consistency.

3️⃣ Isolation

- Each transaction runs as if it is the **only one** on the system.
- Prevents interference between concurrent transactions.

Example:

Two users booking the last ticket at the same time → isolation avoids double booking.

Isolation Levels:

- Read Uncommitted
- Read Committed
- Repeatable Read
- Serializable

4️⃣ Durability

- Once a transaction is committed, **it will survive even system failure**.
- Data is written to non-volatile storage (logs, files).

Example:

After committing a bank transfer, system crash → the transfer still remains.

📌 Summary Table

Property	Meaning
A – Atomicity	All or none
C – Consistency	Maintains rules & constraints
I – Isolation	No interference among transactions
D – Durability	Changes survive crashes

🏛️ Database Instance vs Schema

1 Database Schema (Structure / Blueprint)

- Logical design of the database.
- Describes tables, columns, data types, constraints, relationships.
- **Does not change frequently.**

Example:

pgsql Copy code

```
Student(id INT, name VARCHAR(30), age INT)
```

This structure is the **schema**.

Schema = **Metadata** = Definition of how data will be stored.

---

2 Database Instance (Snapshot of Data)

- The **actual data** stored in the database at a given time.
- Instance changes frequently (insert/update/delete).

Example:

At 10 AM:

Copy code

```
1, Rohan, 20
2, Neha, 22
```

At 2 PM (after updates):

Copy code

```
1, Rohan, 20
2, Neha, 22
3, Aditya, 21
```

Both are different **instances** of the same schema.

📌 Quick Comparison Table:

Feature	Schema	Instance
Meaning	Structure	Actual Data
Changes	Rare	Very frequent
Type	Logical	Physical
Example	Table definition	Rows in the table

If you want, I can also give **practice questions** or **MCQs** on ACID and Schema/Instance.

MySQLStorage Engines (InnoDB, MyISAM and others),

Here is the complete, exam-focused explanation of **MySQL Storage Engines** (InnoDB, MyISAM, and others) with comparison tables and key points.

⭐ MySQL Storage Engines – Overview

A **storage engine** is the internal component of MySQL that handles:

- How data is stored on disk
- How indexes work
- How transactions are processed
- Locking and concurrency behavior

MySQL allows choosing different storage engines **per table** using:

sql Copy code

```
ENGINE = InnoDB;
```

🟩 1. InnoDB (Default Storage Engine)

Best for: OLTP systems, banking, e-commerce, transactions

✅ Features:

- **ACID compliant**
- **Transaction support** (COMMIT, ROLLBACK, SAVEPOINT)
- **Row-level locking**
- **Foreign keys support**
- MVCC (Multi-Version Concurrency Control)
- Crash recovery
- Better performance for **read/write operations**

✔️ Internally:

- Stores data in **clustered indexes** (Primary key order)
- Uses **redo logs**, **undo logs**

🟢 Use When:

- You need reliability, consistency
- High concurrency
- Complex relational constraints

🟦 2. MyISAM

Best for: Read-heavy workloads, analytics, logging

❌ No transaction support

❌ No foreign keys

❌ No crash recovery

✅ Features:

- Fast **read performance**
- **Table-level locking**
- Smaller storage footprint compared to InnoDB

- Supports **full-text indexing** (InnoDB supports from MySQL 5.6+)

🟢 Use When:

- Mostly SELECT queries
- No need for transactions
- High-speed reading is required
- Simple tables, no relationships

🟥 3. MEMORY (HEAP)

Data stored in RAM → Extremely fast

Features:

- Table lives in memory; data lost if server restarts
- Table-level locking
- Good for temporary/lookup tables

Use When:

- Caching queries
- Session management
- Temporary calculations

🟪 4. CSV Storage Engine

- Stores data in **comma-separated text files**
- Easy for external apps to read/write
- No indexing or transactions
- Very slow

Use When:

- Export/import simple data
- Working with spreadsheets (Excel/Google Sheets)

🟧 5. ARCHIVE Storage Engine

- Highly compressed storage
- Allows **INSERT** and **SELECT** only (no UPDATE/DELETE)
- Used for storing large logs or data warehouses

Use When:

- Storing audit logs
- Historical data

🟫 6. BLACKHOLE Storage Engine

- "/dev/null" engine → accepts data but **stores nothing**
- Used for replication (master writes, slaves process)

Use When:

- Logging in replication setups
- Auditing query flow without storing data

🟤 7. FEDERATED Storage Engine

- Allows creating a table that connects to a table in another MySQL server
- No local data storage

Use When:

- Distributed database architecture
- Remote data access

🔥 8. NDB (MySQL Cluster Engine)

- Used for MySQL Cluster
- Provides high availability, auto-sharding

Features:

- Real-time performance
- Multi-master distributed system

- In-memory by default

Use When:

- Telecom
- Banking workloads
- Highly available clustered systems

📄 Comparison: InnoDB vs MyISAM

Feature	InnoDB	MyISAM
Transactions	Yes	No
Foreign Keys	Yes	No
Locking	Row-level	Table-level
Crash Recovery	Yes	No
Speed	Fast for R/W	Fast for reads
Storage	Larger	Smaller
Default Engine	Yes	No

📝 How to Check Storage Engine

sql Copy code

```sql
SHOW TABLE STATUS LIKE 'table_name';
```

📝 How to set storage engine while creating table

sql Copy code

```
CREATE TABLE students (
    id INT PRIMARY KEY,
    name VARCHAR(50)
) ENGINE=InnoDB;
```

📝 Change storage engine of an existing table

sql Copy code

```
ALTER TABLE students ENGINE=MyISAM;
```

If you want, I can give **MCQs**, **short notes**, or a **full exam revision sheet** on MySQL storage engines.

Introduction to MySQL Programming, Use of MySQL Programs,

Below is a clear, exam-oriented explanation for **Introduction to MySQL Programming** and **Use of MySQL Programs**.

⭐ Introduction to MySQL Programming

MySQL Programming refers to writing commands, scripts, and applications that interact with a MySQL database.
It includes:

✔ Writing SQL statements

- CREATE, INSERT, UPDATE, DELETE
- SELECT queries

- Functions, joins, triggers, views, procedures

✔️ Writing MySQL programs

Programs or scripts written in:

- **MySQL Shell**
- **MySQL Monitor (Command-Line Client)**
- **MySQL Workbench**
- External languages that connect to MySQL using APIs:
  - Java (JDBC)
  - Python (mysql-connector)
  - PHP (PDO)
  - C/C++
  - Node.js
  - .NET (C#)

Goal:
To communicate with the database, execute SQL operations, and build data-driven applications.

⭐ Why MySQL Programming is Important?

1. To interact with databases programmatically
2. Automate database tasks
3. Build applications that store and retrieve data
4. Implement business logic using stored procedures
5. Control data access, security, and transactions

⭐ Use of MySQL Programs

MySQL provides several built-in client programs to interact with the server.
Most commonly used MySQL programs are:

🟩 1. MySQL Server Program (`mysqld`)

- The main database server
- Manages all databases, users, connections
- Stores and retrieves data
- Runs as a background service

🟦 2. MySQL Client Program (`mysql`) — MySQL Monitor

This is the **command line tool** used to communicate with MySQL Server.

✔ You can:

- Login to database
- Run SQL commands
- Execute scripts
- Import & export data
- Manage tables

Example:

bash Copy code

```
mysql -u root -p
```

🟥 3. MySQL Workbench

A graphical (GUI) tool used for:

- Writing & running SQL queries
- Designing ER diagrams
- Managing users
- Monitoring server performance
- Database administration

It is user-friendly and ideal for beginners.

🟧 4. MySQL Shell (`mysqlsh`)

A modern MySQL client that supports multiple modes:

Mode	Language
SQL	Standard SQL commands
JavaScript	Scripting automation
Python	Scripting automation

Features:

- Scripting support
- Admin tasks (backup, replication)
- Works with MySQL X DevAPI
- Better for developers than command-line client

🟫 5. MySQL Utilities & Tools

Tool	Purpose
`mysqldump`	Backup & export databases
`mysqlimport`	Import data
`mysqladmin`	Server administration
`mysqlshow`	Display database/table information
`mysqlbinlog`	Read binary logs
`mysql_upgrade`	Upgrade data dictionary on new versions

🟪 6. API/Connector Programs

Used when writing applications with MySQL:

Language	Connector
Java	JDBC

Language	Connector
Python	mysql-connector
PHP	PDO_MySQL
Node.js	mysql2
C/C++	libmysqlclient

They allow programs to:

- Connect to MySQL
- Run queries
- Handle transactions
- Process results

🌟 Summary

MySQL Programming Includes:

- Writing SQL queries
- Writing scripts (Python, JavaScript, SQL Shell)
- Using MySQL programs to manage data

Use of MySQL Programs:

- MySQL Server → stores data
- MySQL Client → run SQL commands
- MySQL Workbench → GUI interface
- MySQL Shell → advanced scripting
- mysqldump/mysqlimport/mysqladmin → backup & admin tasks
- API connectors → communicate through programming languages

If you want, I can also provide:

✅ Very short answers (2–3 marks)
✅ Long answers (6–8 marks)

✅ Exam revision notes

✅ Multiple-choice questions (MCQs) on this topic

Introduction to Stored Procedures, Benefits of Stored Procedures

Below is a clean, simple, and exam-focused explanation of **Stored Procedures** and their **Benefits**, perfect for theory, viva, and written exams.

⭐ Introduction to Stored Procedures

A **Stored Procedure** is a **precompiled group of SQL statements** that are stored and executed on the MySQL server.

In simple terms:
➡️ A stored procedure is a **function-like program** inside the database that performs a specific task.

✔️ Key Points:

- Stored in the database server
- Executes with a single call
- Can include SQL statements, loops, conditions, variables
- Can accept **parameters (IN, OUT, INOUT)**
- Increases speed and reusability

📝 Example of a Simple Stored Procedure

sql Copy code

```
DELIMITER $$
```

```
CREATE PROCEDURE getAllStudents()
BEGIN
    SELECT * FROM students;
END $$

DELIMITER ;
```

To execute:

sql Copy code

```
CALL getAllStudents();
```

⭐ Benefits of Stored Procedures

Stored procedures provide multiple performance, security, and development advantages.

🟩 1. Performance Improvement

- SQL statements inside the procedure are **compiled and stored** the first time they run.
- Later executions are faster.

🟦 2. Reusability

- Common operations (like insert, update, calculations) can be reused many times.
- Reduces duplicate SQL code in applications.

🟥 3. Reduced Network Traffic

Instead of sending multiple SQL queries from the application to the server, you send **one CALL statement**.

Example:

sql Copy code

```
CALL updateSalary(101, 5000);
```

🟧 4. Better Security

- Access to tables can be restricted.
- Users can be allowed to execute the procedure **without giving permission** to underlying tables.
- Helps in enforcing business rules.

Example:

sql Copy code

```
GRANT EXECUTE ON PROCEDURE updateSalary TO 'user1'@'localhost';
```

🟫 5. Business Logic on Server Side

- Important rules and validations stored on server
- Application becomes simpler
- Centralized logic → easy to maintain

🟪 6. Reduces Errors

- Tested once → works everywhere
- Eliminates repetitive coding mistakes
- Ensures consistent output

🟩 7. Transaction Control

Stored procedures can use:

- COMMIT
- ROLLBACK
- SAVEPOINT

Good for:

- Banking
- Inventory
- Sales systems

🟦 8. Easier Maintenance

- If logic changes, update only the procedure
- No need to change application code

🟥 9. Supports Modularity

Breaks a complex task into smaller procedures (like functions in programming).

⭐ Summary (Perfect for 2–5 Marks)

Stored Procedure

A stored procedure is a precompiled set of SQL statements stored in the database server, used to perform a specific task. It supports parameters, transaction control, and can contain programming logic.

Benefits

1. Faster performance
2. Reusable code
3. Reduced network load
4. Higher security
5. Centralized business logic
6. Reduces errors
7. Supports transactions
8. Easier maintenance
9. Modular programming

If you want next topics like:

✅ Functions
✅ Triggers
✅ Cursors
— just tell me!

Procedure Parameters (IN, OUT and INOUT).

Here is a clear, complete, and exam-friendly explanation of **Procedure Parameters in MySQL**: **IN, OUT, and INOUT**, with examples and notes.

⭐ Procedure Parameters in MySQL

Stored procedures in MySQL can take parameters to pass values **into** or **out of** the procedure.

There are **three types** of parameters:

1. **IN parameter**
2. **OUT parameter**
3. **INOUT parameter**

These help procedures act like functions and perform dynamic operations.

🟩 1. IN Parameter

✔ **Used to pass a value into the procedure.**

✔ **Default type (if not specified).**

✔ **Value is read-only inside the procedure.**

✔ Cannot be changed or returned to the calling program.

🔹 Example: IN Parameter

sql Copy code

```
DELIMITER $$

CREATE PROCEDURE getStudent(IN stu_id INT)
BEGIN
    SELECT * FROM students WHERE id = stu_id;
END $$

DELIMITER ;
```

Call it:

sql Copy code

```
CALL getStudent(5);
```

🟦 2. OUT Parameter

✔ Used to return a value from the procedure.

✔ Value can be modified inside the procedure.

✔ Caller receives the updated value.

🔹 Example: OUT Parameter

sql Copy code

```
DELIMITER $$

CREATE PROCEDURE getTotalStudents(OUT total INT)
BEGIN
    SELECT COUNT(*) INTO total FROM students;
END $$
```

```
DELIMITER ;
```

Call it:

sql Copy code

```
CALL getTotalStudents(@count);
SELECT @count;
```

🟥 3. INOUT Parameter

✔ **Works as both input and output**

✔ **Caller sends a value, procedure modifies it, and returns updated value.**

🔹 **Example: INOUT Parameter**

sql Copy code

```
DELIMITER $$

CREATE PROCEDURE increaseSalary(INOUT salary INT)
BEGIN
    SET salary = salary + 5000;
END $$

DELIMITER ;
```

Call it:

sql Copy code

```
SET @sal = 20000;
CALL increaseSalary(@sal);
SELECT @sal;   -- Result: 25000
```

⭐ Comparison Table

Parameter Type	Input Allowed	Output Allowed	Description
IN	✔ Yes	❌ No	Pass value into procedure
OUT	❌ No	✔ Yes	Return value from procedure
INOUT	✔ Yes	✔ Yes	Pass value and return updated value

📝 Perfect Exam Short Notes

IN Parameter:

Used to send a value to the procedure. Value cannot be changed or returned.

OUT Parameter:

Used to return a value from the procedure using `INTO` or `SET`. Cannot take input.

INOUT Parameter:

Acts as both input and output. Value sent by caller is modified and returned.

If you want, I can explain:

✅ Stored Functions
✅ Cursors
✅ Triggers
✅ Exception Handling in Stored Procedures

Just tell me!

Flow Control Statements (LOOP, WHILE and REPEAT)

Below is a clear, exam-oriented explanation of **Flow Control Statements in MySQL Stored Programs**: **LOOP**, **WHILE**, and **REPEAT** — with syntax and examples.

⭐ Flow Control Statements in MySQL

MySQL Stored Procedures allow procedural programming using control-flow statements. The most commonly used looping constructs are:

1. **LOOP**
2. **WHILE loop**
3. **REPEAT loop**

They help execute a block of statements multiple times.

🟩 1. LOOP Statement

✔ **Runs repeatedly until you explicitly exit it.**

✔ **Must use `LEAVE` to break the loop.**

Syntax:

sql Copy code

```
[loop_label]: LOOP
    -- statements
    LEAVE loop_label;   -- exit condition
END LOOP;
```

Example: Simple LOOP with LEAVE

sql Copy code

```
DELIMITER $$

CREATE PROCEDURE loopExample()
BEGIN
```

```
    DECLARE x INT DEFAULT 1;

    myLoop: LOOP
        SET x = x + 1;
        IF x > 5 THEN
            LEAVE myLoop;    -- exit loop
        END IF;
    END LOOP;

    SELECT x;
END $$

DELIMITER ;
```

🟦 2. WHILE Loop

✔ **Condition is checked before entering the loop.**

✔ **Runs as long as condition is true.**

Syntax:

sql Copy code

```
WHILE condition DO
    -- statements
END WHILE;
```

Example: WHILE Loop

sql Copy code

```
DELIMITER $$

CREATE PROCEDURE whileExample()
BEGIN
    DECLARE i INT DEFAULT 1;

    WHILE i <= 5 DO
```

```
        INSERT INTO log_table VALUES (i);
        SET i = i + 1;
    END WHILE;
END $$

DELIMITER ;
```

🟥 3. REPEAT Loop

✔ Condition is checked after executing the loop body.

✔ Runs at least once (similar to do-while loop in programming).

✔ Loop ends when the condition becomes true.

Syntax:

sql Copy code

```
REPEAT
    -- statements
UNTIL condition END REPEAT;
```

Example: REPEAT Loop

sql Copy code

```
DELIMITER $$

CREATE PROCEDURE repeatExample()
BEGIN
    DECLARE n INT DEFAULT 1;

    REPEAT
        SET n = n + 1;
    UNTIL n > 5
    END REPEAT;

    SELECT n;
```

```
END $$

DELIMITER ;
```

⭐ Comparison Table (Perfect for Viva & Exams)

Feature	LOOP	WHILE	REPEAT
Condition check	No automatic check	Before loop	After loop
Must exit manually?	✔ Yes (using LEAVE)	❌ No	❌ No
Runs at least once?	❌ No	❌ No	✔ Yes
Best use	Complex exit conditions	Known number of iterations	Do-while behavior

---

⭐ Short Notes (2-3 Marks)

LOOP:

A basic loop structure that repeats indefinitely until a `LEAVE` statement exits it. Useful for complex exit conditions.

WHILE:

A pre-test loop that executes as long as the condition remains true.

REPEAT:

A post-test loop that executes at least once and stops when the `UNTIL` condition becomes true.

If you want, I can also explain:

✅ IF / CASE Statements
✅ Cursors

✅ Handlers (Exception handling)

— Just tell me!

Using above statements in Stored Procedures/Functions

Here is a clear, exam-ready explanation of **using Flow Control Statements (LOOP, WHILE, REPEAT, IF, CASE)** inside **Stored Procedures and Stored Functions** — with syntax and complete examples.

⭐ Using Flow Control Statements in Stored Procedures & Functions

MySQL Stored Procedures and Functions support procedural programming using:

- IF / ELSEIF / ELSE
- CASE
- LOOP
- WHILE
- REPEAT
- LEAVE / ITERATE
- Variables

These help create logic similar to programming languages.

🟩 1. Using IF / ELSEIF / ELSE in Procedures

Syntax

sql Copy code

```
IF condition THEN
    statements;
ELSEIF condition THEN
    statements;
ELSE
    statements;
END IF;
```

Example

sql Copy code

```
DELIMITER $$

CREATE PROCEDURE gradeCheck(IN marks INT, OUT grade CHAR(1))
BEGIN
    IF marks >= 90 THEN
        SET grade = 'A';
    ELSEIF marks >= 75 THEN
        SET grade = 'B';
    ELSEIF marks >= 50 THEN
        SET grade = 'C';
    ELSE
        SET grade = 'F';
    END IF;
END $$

DELIMITER ;
```

🟦 2. Using CASE in Stored Procedures

Syntax

sql Copy code

```
CASE
    WHEN condition1 THEN statements;
    WHEN condition2 THEN statements;
    ELSE statements;
END CASE;
```

Example

sql Copy code

```
DELIMITER $$

CREATE PROCEDURE dayName(IN d INT, OUT name VARCHAR(10))
BEGIN
    CASE d
        WHEN 1 THEN SET name = 'Monday';
        WHEN 2 THEN SET name = 'Tuesday';
        WHEN 3 THEN SET name = 'Wednesday';
        ELSE SET name = 'Invalid';
    END CASE;
END $$

DELIMITER ;
```

🟥 3. Using WHILE Loop in Procedures

✔ Condition checked before loop begins

Example

sql Copy code

```
DELIMITER $$

CREATE PROCEDURE whileLoopDemo()
```

```
BEGIN
    DECLARE i INT DEFAULT 1;

    WHILE i <= 5 DO
        INSERT INTO demo VALUES(i);
        SET i = i + 1;
    END WHILE;
END $$

DELIMITER ;
```

🟧 4. Using REPEAT Loop in Procedures

✔️ Condition checked after loop executes (do-while)

Example

sql Copy code

```
DELIMITER $$

CREATE PROCEDURE repeatLoopDemo()
BEGIN
    DECLARE x INT DEFAULT 1;

    REPEAT
        INSERT INTO test VALUES(x);
        SET x = x + 1;
    UNTIL x > 5
    END REPEAT;

END $$

DELIMITER ;
```

--

🟫 5. Using LOOP and LEAVE in Procedures

✔ LOOP runs indefinitely until `LEAVE` is used.

Example

sql Copy code

```sql
DELIMITER $$

CREATE PROCEDURE loopDemo()
BEGIN
    DECLARE n INT DEFAULT 1;

    myLoop: LOOP
        SET n = n + 1;

        IF n > 5 THEN
            LEAVE myLoop;   -- exit the loop
        END IF;

    END LOOP;

    SELECT n AS result;
END $$

DELIMITER ;
```

--

🟣 6. Using ITERATE (Continue) Inside LOOP

✔ Used to skip remaining statements and jump to next iteration.

Example

sql Copy code

```sql
DELIMITER $$

CREATE PROCEDURE iterateDemo()
BEGIN
    DECLARE i INT DEFAULT 0;

    loop1: LOOP
        SET i = i + 1;

        IF i = 3 THEN
            ITERATE loop1;  -- skip 3
        END IF;

        INSERT INTO numbers VALUES(i);

        IF i >= 5 THEN
            LEAVE loop1;
        END IF;
    END LOOP;
END $$

DELIMITER ;
```

⭐ Using Flow Control in Stored Functions

Stored functions work similarly but must **return a value**.

Example with WHILE loop in a FUNCTION

sql Copy code

```sql
DELIMITER $$

CREATE FUNCTION factorial(n INT)
```

```
RETURNS INT
BEGIN
    DECLARE result INT DEFAULT 1;
    DECLARE i INT DEFAULT 1;

    WHILE i <= n DO
        SET result = result * i;
        SET i = i + 1;
    END WHILE;

    RETURN result;
END $$

DELIMITER ;
```

Usage:

sql Copy code

```
SELECT factorial(5);
```

🌟 Summary (Perfect for Exams)

Stored Procedures/Functions support:

- **IF**, **CASE** → decision-making
- **WHILE**, **REPEAT**, **LOOP** → looping
- **LEAVE**, **ITERATE** → loop control
- Can include variables, conditions, SQL statements

Used for:

- Business logic
- Validations
- Repetitive tasks
- Calculations inside database

If you want next topics:

✅ Cursors
✅ Handlers (Exception Handling)
✅ Triggers

Just tell me!

Conditional Statements (IF, IF-ELSE-THEN, SWITCH CASE)

Below is a **simple, clean, and exam-oriented** explanation of **Conditional Statements in MySQL** — **IF**, **IF-THEN-ELSE**, and **CASE / SWITCH CASE —** with syntax and examples.

⭐ Conditional Statements in MySQL

Conditional statements allow MySQL stored programs (Procedures/Functions) to make decisions based on conditions—similar to programming languages.

MySQL provides:

1. **IF (simple IF)**
2. **IF-THEN-ELSE / IF-ELSEIF-ELSE**
3. **CASE (acts like SWITCH CASE)**

🟩 1. IF Statement (Simple IF)

Used when you want to execute a statement *only if* a condition is TRUE.

✔ Syntax:

sql Copy code

```
IF condition THEN
    statements;
```

```
END IF;
```

✔ Example:

sql Copy code

```
IF marks >= 40 THEN
    SET result = 'Pass';
END IF;
```

🟦 2. IF–THEN–ELSE Statement

Used when you want to choose between **two** possible actions.

✔ Syntax:

sql Copy code

```
IF condition THEN
    statements;
ELSE
    statements;
END IF;
```

✔ Example:

sql Copy code

```
IF age >= 18 THEN
    SET status = 'Adult';
ELSE
    SET status = 'Minor';
END IF;
```

🟥 3. IF–ELSEIF–ELSE (Multiple Conditions)

Used when you have **more than two** possibilities.

✔ Syntax:

sql Copy code

```sql
IF condition1 THEN
    statements;
ELSEIF condition2 THEN
    statements;
ELSEIF condition3 THEN
    statements;
ELSE
    statements;
END IF;
```

✔ Example:

sql Copy code

```sql
IF marks >= 90 THEN
    SET grade = 'A';
ELSEIF marks >= 75 THEN
    SET grade = 'B';
ELSEIF marks >= 50 THEN
    SET grade = 'C';
ELSE
    SET grade = 'F';
END IF;
```

🟧 4. CASE Statement (SWITCH CASE Equivalent)

`CASE` is used when you want to test multiple conditions—similar to **switch-case** in programming.

⭐ Type 1: CASE - Value-based (Switch Case style)

✔ Syntax:

sql

```
CASE value
    WHEN option1 THEN statements;
    WHEN option2 THEN statements;
    ELSE statements;
END CASE;
```

✔ Example:

sql Copy code

```
CASE day_num
    WHEN 1 THEN SET day_name = 'Monday';
    WHEN 2 THEN SET day_name = 'Tuesday';
    WHEN 3 THEN SET day_name = 'Wednesday';
    ELSE SET day_name = 'Invalid';
END CASE;
```

⭐ Type 2: CASE - Condition-based

Useful when each case has a separate condition.

✔ Syntax:

sql Copy code

```
CASE
    WHEN condition1 THEN statements;
    WHEN condition2 THEN statements;
    ELSE statements;
END CASE;
```

✔ Example:

sql Copy code

```
CASE
    WHEN salary > 50000 THEN SET tax = 'High';
    WHEN salary > 20000 THEN SET tax = 'Medium';
```

```
    ELSE SET tax = 'Low';
END CASE;
```

⭐ Comparison Table

Statement	Supports Multiple Conditions?	Similar To
IF	❌ Mostly simple checks	if()
IF-ELSE	✔ Two options	if-else
IF-ELSEIF-ELSE	✔✔ Multiple checks	else-if ladder
CASE	✔✔✔ Multiple cases, cleaner	switch-case

⭐ Usage Inside Stored Procedures

Example combining all:

sql Copy code

```sql
DELIMITER $$

CREATE PROCEDURE checkNumber(IN num INT, OUT result VARCHAR(20))
BEGIN
    IF num = 0 THEN
        SET result = 'Zero';

    ELSEIF num > 0 THEN
        CASE
            WHEN num < 10 THEN SET result = 'Positive Single Digit';
            WHEN num < 100 THEN SET result = 'Positive Two Digit';
            ELSE SET result = 'Big Positive Number';
        END CASE;

    ELSE
        SET result = 'Negative';
    END IF;
```

```
END $$

DELIMITER ;
```

🌟 Summary (Perfect for Exams)

IF

Executes statements only when a condition is true.

IF-ELSE

Executes one block if condition is true, otherwise another block.

IF-ELSEIF-ELSE

Executes one of many possible blocks based on multiple conditions.

CASE (Switch Case)

Cleaner alternative for multiple conditions; supports value-based and condition-based checks.

If you want, I can explain:
✅ Loops (LOOP, WHILE, REPEAT)
✅ Cursors
✅ Triggers
Just say the topic!

Example of each type of statement

Below are **clear and simple examples** of each conditional statement in **MySQL Stored Procedures/Functions**:

✅ 1. IF Statement (Single Condition)

✔ Example

sql Copy code

```sql
DELIMITER $$

CREATE PROCEDURE check_age(IN age INT)
BEGIN
    IF age >= 18 THEN
        SELECT 'Eligible to vote' AS result;
    END IF;
END $$

DELIMITER ;
```

✅ 2. IF-ELSE Statement

✔ Example

sql Copy code

```sql
DELIMITER $$

CREATE PROCEDURE check_number(IN num INT)
BEGIN
    IF num > 0 THEN
        SELECT 'Positive Number' AS result;
    ELSE
        SELECT 'Negative or Zero' AS result;
    END IF;
END $$

DELIMITER ;
```

✅ 3. IF – ELSEIF – ELSE Chain

✔ Example

sql Copy code

```sql
DELIMITER $$

CREATE PROCEDURE categorize_marks(IN marks INT)
BEGIN
    IF marks >= 75 THEN
        SELECT 'Distinction' AS result;
    ELSEIF marks >= 60 THEN
        SELECT 'First Class' AS result;
    ELSEIF marks >= 40 THEN
        SELECT 'Pass' AS result;
    ELSE
        SELECT 'Fail' AS result;
    END IF;
END $$

DELIMITER ;
```

✅ 4. CASE (Switch Case Equivalent)

MySQL uses `CASE` instead of `SWITCH`.

✔ Example

sql Copy code

```sql
DELIMITER $$

CREATE PROCEDURE day_name(IN day_num INT)
BEGIN
    CASE day_num
        WHEN 1 THEN SELECT 'Monday';
        WHEN 2 THEN SELECT 'Tuesday';
        WHEN 3 THEN SELECT 'Wednesday';
```

```
        WHEN 4 THEN SELECT 'Thursday';
        WHEN 5 THEN SELECT 'Friday';
        WHEN 6 THEN SELECT 'Saturday';
        WHEN 7 THEN SELECT 'Sunday';
        ELSE SELECT 'Invalid Day';
    END CASE;
END $$

DELIMITER ;
```

💡 Summary Table

Statement Type	Purpose	Sample Use
IF	Check a single condition	IF age ≥ 18
IF-ELSE	Two conditions	Positive / Negative
IF-ELSEIF-ELSE	Multiple conditions	Marks grade
CASE	Multiple values of a variable	Day number to name

If you want, I can also give examples using **functions**, **loops**, **cursors**, or real-world stored procedure programs.

Loop constructs (ITERATE, LEAVE)

Here are the **Loop constructs in MySQL** using **ITERATE** and **LEAVE**, explained with **simple, clear examples**.

✅ 1. LEAVE Statement (Exit from loop)

`LEAVE` works like **break** in other languages.

✔️ Example: Exit loop when counter reaches 5

sql Copy code

```sql
DELIMITER $$

CREATE PROCEDURE demo_leave()
BEGIN
    DECLARE counter INT DEFAULT 1;

    loop_label: LOOP
        IF counter > 5 THEN
            LEAVE loop_label;   -- Exit the loop
        END IF;

        SELECT CONCAT('Counter = ', counter);
        SET counter = counter + 1;
    END LOOP loop_label;
END $$

DELIMITER ;
```

🔍 Output

Counter = 1

Counter = 2

Counter = 3

Counter = 4

Counter = 5

When `counter > 5`, loop stops.

✅ 2. ITERATE Statement (Skip iteration — similar to continue)

`ITERATE` means **go to the next iteration of loop**.

✔ Example: Skip printing odd numbers

sql Copy code

```sql
DELIMITER $$

CREATE PROCEDURE demo_iterate()
BEGIN
    DECLARE num INT DEFAULT 1;

    loop_label: LOOP
        IF num > 10 THEN
            LEAVE loop_label;   -- Exit the loop
        END IF;

        IF (num % 2) = 1 THEN
            SET num = num + 1;
            ITERATE loop_label;   -- Skip this iteration
        END IF;

        SELECT CONCAT('Even Number = ', num);
        SET num = num + 1;
    END LOOP loop_label;
END $$

DELIMITER ;
```

🔍 Output

Even Number = 2
Even Number = 4
Even Number = 6
Even Number = 8
Even Number = 10

Odd numbers are **skipped** using `ITERATE`.

🎯 Summary

Statement	Purpose	Equivalent In Other Languages
LEAVE	Exit loop completely	`break`
ITERATE	Skip current iteration and continue	`continue`

If you want, I can also provide **examples using LOOP + WHILE + REPEAT combined with LEAVE/ITERATE**, or real-world stored procedures.

MySQLBuilt-in functions (string, numeric, date etc.)

Below is a **complete, easy-to-understand list of MySQL Built-in Functions** with **String, Numeric, Date/Time, Aggregate, and Conversion** functions — with examples.

⭐ MySQL Built-in Functions

✔️ 1. String Functions

🔹 1. CONCAT() – Combine strings

sql Copy code

```sql
SELECT CONCAT('Hello ', 'World');
```

🔹 2. LENGTH() – Length of string (bytes)

sql Copy code

```sql
SELECT LENGTH('MySQL');
```

3. LOWER() / UPPER()

sql Copy code

```sql
SELECT LOWER('HELLO'), UPPER('hello');
```

4. SUBSTRING() – Extract part of string

sql Copy code

```sql
SELECT SUBSTRING('Database', 1, 4);   -- Data
```

5. REPLACE()

sql Copy code

```sql
SELECT REPLACE('Hello World', 'World', 'MySQL');
```

6. TRIM(), LTRIM(), RTRIM()

sql Copy code

```sql
SELECT TRIM('   Hi   ');
```

7. LEFT() / RIGHT()

sql Copy code

```sql
SELECT LEFT('MySQL', 2);   -- My
SELECT RIGHT('MySQL', 2);  -- SQL
```

8. INSTR() – Find string position

sql Copy code

```sql
SELECT INSTR('MySQL', 'SQL');  -- 3
```

9. REVERSE()

sql Copy code

```sql
SELECT REVERSE('ABC');    -- CBA
```

10. LPAD() / RPAD()

sql Copy code

```sql
SELECT LPAD('7', 3, '0');  -- 007
```

---

2. Numeric Functions

1. ABS() – Absolute value

sql Copy code

```sql
SELECT ABS(-10);    -- 10
```

2. CEIL() / FLOOR()

sql Copy code

```sql
SELECT CEIL(5.2);  -- 6
SELECT FLOOR(5.8); -- 5
```

3. ROUND()

sql Copy code

```sql
SELECT ROUND(5.678, 2);  -- 5.68
```

4. MOD()

sql Copy code

```sql
SELECT MOD(10, 3);  -- 1
```

5. POWER()

sql Copy code

```sql
SELECT POWER(2, 3); -- 8
```

6. SQRT()

sql Copy code

```sql
SELECT SQRT(25);   -- 5
```

7. RAND()

sql Copy code

```sql
SELECT RAND();  -- random number 0-1
```

---

3. Date & Time Functions

1. CURRENT_DATE(), CURRENT_TIME(), NOW()

sql Copy code

```sql
SELECT CURRENT_DATE(), CURRENT_TIME(), NOW();
```

2. DATE() / TIME()

sql Copy code

```sql
SELECT DATE(NOW());  -- Extract date
SELECT TIME(NOW());  -- Extract time
```

3. YEAR(), MONTH(), DAY()

sql Copy code

```sql
SELECT YEAR(NOW()), MONTH(NOW()), DAY(NOW());
```

4. DATE_ADD() / DATE_SUB()

sql Copy code

```sql
SELECT DATE_ADD('2025-01-01', INTERVAL 10 DAY);
```

5. DATEDIFF()

sql Copy code

```sql
SELECT DATEDIFF('2025-12-31','2025-01-01');  -- 364 days
```

6. DAYNAME(), MONTHNAME()

sql Copy code

```sql
SELECT DAYNAME('2025-01-10');  -- Friday
```

7. LAST_DAY()

sql Copy code

```sql
SELECT LAST_DAY('2025-02-10');  -- 2025-02-28
```

✔️ 4. Aggregate Functions

Used with **GROUP BY**.

SUM()

sql Copy code

```sql
SELECT SUM(salary) FROM employees;
```

AVG()

sql Copy code

```sql
SELECT AVG(marks) FROM students;
```

COUNT()

sql Copy code

```sql
SELECT COUNT(*) FROM orders;
```

MAX() / MIN()

sql Copy code

```sql
SELECT MAX(price), MIN(price) FROM products;
```

---

5. Conversion Functions

CAST()

sql Copy code

```sql
SELECT CAST('123' AS SIGNED);  -- converts to number
```

CONVERT()

sql Copy code

```sql
SELECT CONVERT('2025-10-10', DATE);
```

---

6. Conditional Functions

IF()

sql Copy code

```sql
SELECT IF(10 > 5, 'YES', 'NO');
```

🔹 IFNULL()

sql Copy code

```sql
SELECT IFNULL(NULL, 'No value');
```

🔹 NULLIF()

sql Copy code

```sql
SELECT NULLIF(10,10);  -- returns NULL
```

✔️ 7. Math & Miscellaneous Functions

🔹 GREATEST() / LEAST()

sql Copy code

```sql
SELECT GREATEST(10, 25, 7);  -- 25
```

🔹 FORMAT()

sql Copy code

```sql
SELECT FORMAT(12345.6789, 2);  -- 12,345.68
```

🎯 Want a PDF of all functions?

I can generate a **full DBMS Notes PDF** with all these topics if you want!

Cursors (Asensitive, Insensitive, Read only, Nonscrollable)
Cursors example and real time use

Below is a **clear, exam-oriented explanation of MySQL Cursors** including **types, syntax, examples, and real-time use case**.

⭐ Cursors in MySQL

A **cursor** is a database object used to **retrieve rows one-by-one** from a result set.

Useful when:

- ✔ You must process **each row individually**
- ✔ You need **row-by-row logic**, which SQL normally avoids

⭐ Types of Cursors in MySQL

MySQL supports limited cursor features (only inside Stored Procedures).

✔ 1. Asensitive Cursor

- The cursor **returns actual data from the table**, not a copy.
- The result may **change if underlying table changes** (not guaranteed stable).
- MySQL **default cursor type**.

✔ 2. Insensitive Cursor *(Not fully supported in MySQL)*

- Returns a **temporary copy** of data.
- Changes in base table after cursor opens **do NOT affect the cursor**.

✔ 3. Read-only Cursor

- You **cannot update** the rows fetched using the cursor.
- MySQL cursors are always **read-only**.

✔ 4. Nonscrollable Cursor

- You can only **move forward** row-by-row.
- Cannot go backward or jump.
- MySQL supports only **FORWARD ONLY** cursors.

⭐ Summary Table

Cursor Type	Supported?	Meaning
Asensitive	✔ Yes	Returns actual table data (default)
Insensitive	✖ No	Returns snapshot copy (not supported)
Read-only	✔ Yes	Cannot update fetched rows
Nonscrollable	✔ Yes	Process rows only in forward direction

⭐ MySQL Cursor Syntax

Inside stored procedure:

1️⃣ Declare cursor
2️⃣ Open cursor
3️⃣ Fetch rows
4️⃣ Close cursor

⭐ Cursor Example (Simple)

Fetch all employee names one-by-one.

sql Copy code

```sql
DELIMITER $$

CREATE PROCEDURE cursor_example()
BEGIN
    DECLARE done INT DEFAULT FALSE;
    DECLARE emp_name VARCHAR(50);

    -- Step 1: Declare cursor
    DECLARE emp_cursor CURSOR FOR
        SELECT name FROM employees;
```

```
    -- Step 2: Declare handler for end-of-data
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;

    -- Step 3: Open cursor
    OPEN emp_cursor;

    read_loop: LOOP
        FETCH emp_cursor INTO emp_name;

        IF done THEN
            LEAVE read_loop;
        END IF;

        SELECT CONCAT('Employee: ', emp_name);
    END LOOP;

    -- Step 4: Close cursor
    CLOSE emp_cursor;
END $$

DELIMITER ;
```

⭐ OUTPUT

makefile Copy code

```
Employee: Ramesh
Employee: Suresh
Employee: Anita
Employee: Priya
```

⭐ Real-Time Example: Calculate Bonus for Each Employee

Scenario:

You want to add **10% bonus** to every employee whose salary < 50,000.

Cursor is useful because this is **row-by-row processing** with conditions.

Stored Procedure Using Cursor (Real Use Case)

sql

Copy code

```sql
DELIMITER $$

CREATE PROCEDURE update_bonus()
BEGIN
    DECLARE done INT DEFAULT FALSE;
    DECLARE eid INT;
    DECLARE esalary DECIMAL(10,2);
    DECLARE new_bonus DECIMAL(10,2);

    DECLARE cur CURSOR FOR
        SELECT id, salary FROM employees;

    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;

    OPEN cur;

    bonus_loop: LOOP
        FETCH cur INTO eid, esalary;

        IF done THEN
            LEAVE bonus_loop;
        END IF;

        IF esalary < 50000 THEN
            SET new_bonus = esalary * 0.10;

            UPDATE employees
            SET bonus = new_bonus
            WHERE id = eid;
        END IF;

    END LOOP;

    CLOSE cur;
END $$
```

```
DELIMITER ;
```

⭐ Real-Time Use Cases of Cursors

✔ 1. Payroll Processing

Calculate salary, bonus, PF, deductions **row-by-row** for each employee.

✔ 2. Generating Monthly Bills

Utility billing (electricity/water) where each customer's bill is processed one by one.

✔ 3. Data Migration

Copy rows from one table to another with transformation logic.

✔ 4. Sending Notifications / Emails

Loop through users and send emails/messages.

✔ 5. Complex validations

When each row needs custom validation before updating.

🎉 If you want, I can also give you:

✅ Cursor examples using **multiple tables**
✅ A full **DBMS+MySQL Notes PDF**
✅ Cursor interview questions

Just tell me!

Triggers (BEFORE, AFTER), New and Old trigger variables
Trigger Examples and real time use

Below is a **complete, clear explanation of MySQL Triggers**, including **BEFORE & AFTER**, **NEW/OLD variables**, **examples**, and **real-time applications**.

⭐ What is a Trigger?

A **trigger** is a stored program that automatically executes
👉 **before** or **after** an INSERT, UPDATE, or DELETE.

Triggers are used for:

- ✔ Validation
- ✔ Auditing
- ✔ Automatic updates
- ✔ Maintaining logs
- ✔ Restricting invalid data

⭐ Types of Triggers in MySQL

Timing	Event	Trigger Name Example
BEFORE	INSERT	BEFORE INSERT trigger
BEFORE	UPDATE	BEFORE UPDATE trigger
BEFORE	DELETE	BEFORE DELETE trigger
AFTER	INSERT	AFTER INSERT trigger
AFTER	UPDATE	AFTER UPDATE trigger
AFTER	DELETE	AFTER DELETE trigger

⭐ OLD and NEW Variables

Trigger Type	OLD	NEW
INSERT	❌ Not available	✔ NEW – holds new row
UPDATE	✔ OLD – old value	✔ NEW – new value
DELETE	✔ OLD – old value	❌ Not available

⭐ 1. BEFORE INSERT Trigger Example

✔ Use: Validate data BEFORE inserting.

Example: Prevent negative salary

sql Copy code

```
DELIMITER $$

CREATE TRIGGER before_salary_insert
BEFORE INSERT ON employees
FOR EACH ROW
BEGIN
    IF NEW.salary < 0 THEN
        SET NEW.salary = 0;  -- Correct invalid value
    END IF;
END $$

DELIMITER ;
```

✔ If user tries to insert negative salary → it becomes 0.

⭐ 2. BEFORE UPDATE Trigger Example

✔ Use: Modify data BEFORE updating.

Example: Automatically convert names to Proper Case.

sql Copy code

```
DELIMITER $$

CREATE TRIGGER before_name_update
BEFORE UPDATE ON employees
FOR EACH ROW
BEGIN
    SET NEW.name = CONCAT(UPPER(LEFT(NEW.name,1)), LOWER(SUBSTRING(NEW.name,2)));
END $$

DELIMITER ;
```

⭐ 3. AFTER INSERT Trigger Example

✔ Use: Insert record into log-table after successful insert.

Example: Audit insertion into log table.

sql Copy code

```
DELIMITER $$

CREATE TRIGGER after_employee_insert
AFTER INSERT ON employees
FOR EACH ROW
BEGIN
    INSERT INTO employee_log(emp_id, action, action_time)
    VALUES (NEW.id, 'INSERT', NOW());
END $$

DELIMITER ;
```

⭐ 4. AFTER UPDATE Trigger Example

✔ Use: Log changes after update.

Example: Save old vs new salary for auditing.

sql Copy code

```sql
DELIMITER $$

CREATE TRIGGER after_salary_update
AFTER UPDATE ON employees
FOR EACH ROW
BEGIN
    INSERT INTO salary_audit(emp_id, old_salary, new_salary, updated_at)
    VALUES (OLD.id, OLD.salary, NEW.salary, NOW());
END $$

DELIMITER ;
```

⭐ 5. BEFORE DELETE Trigger Example

✔ Use: Restrict deletion based on condition.

Example: Block deleting admin user.

sql Copy code

```sql
DELIMITER $$

CREATE TRIGGER before_employee_delete
BEFORE DELETE ON employees
FOR EACH ROW
BEGIN
    IF OLD.role = 'admin' THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Admin cannot be deleted';
    END IF;
END $$

DELIMITER ;
```

⭐ 6. AFTER DELETE Trigger Example

✔ Use: Log deleted data.

sql Copy code

```sql
DELIMITER $$

CREATE TRIGGER after_employee_delete
AFTER DELETE ON employees
FOR EACH ROW
BEGIN
    INSERT INTO deleted_employees(id, name, salary, deleted_at)
    VALUES (OLD.id, OLD.name, OLD.salary, NOW());
END $$

DELIMITER ;
```

⭐ Real-Time Use Cases of Triggers

✔ 1. Audit Logging

Record when a row is:

- Inserted
- Updated
- Deleted

Used in banks, HR systems, billing systems.

✔ 2. Ensure Data Integrity

- Prevent invalid salary
- Auto-correct wrong inputs
- Restrict deletion of key records

✔ 3. Auto-maintain calculated columns

Example:

Total Amount = quantity × price

Auto-update after update to either.

✔ 4. Maintain History Tables

Keep full history of:

- Previous salaries
- Previous address
- Previous product price

✔ 5. Cascading actions (custom)

If a parent table updated → automatically update child table.

✔ 6. Email/SMS Notification (via log table)

Trigger inserts entry → backend app reads log table → sends notification.

⭐ Want a PDF of all Triggers Notes + Examples?

I can generate it for you!

Error Handling and Exceptions, Types of Handler Actions, How to write Handler
Defining and handling exceptions in Stored Procedures and Functions

Here's a **complete, exam-ready explanation of Error Handling and Exception Handling in MySQL**, including **types of handlers, syntax, and examples**.

⭐ 1. What is Error Handling / Exception Handling?

In MySQL Stored Procedures/Functions, **errors or exceptional conditions** may occur, such as:

- Division by zero
- Duplicate key insertion
- Data not found
- Invalid input

Exception handling allows you to **catch and handle these errors** gracefully instead of letting the procedure fail.

⭐ 2. HANDLER in MySQL

MySQL uses **HANDLER statements** inside stored programs to handle errors/exceptions.

Syntax:

sql Copy code

```
DECLARE handler_type HANDLER FOR condition_value
    statement;
```

- **handler_type**: CONTINUE or EXIT
- **condition_value**: SQLSTATE code, MySQL error code, or `SQLEXCEPTION`, `SQLWARNING`, `NOT FOUND`
- **statement**: action to execute

⭐ 3. Types of Handler Actions

Type	Behavior	Example
CONTINUE HANDLER	After handling, procedure **continues execution**	Skip error, continue next statements
EXIT HANDLER	After handling, procedure **exits current block**	Stop execution of procedure

Type	Behavior	Example
UNDO HANDLER *(Not supported in MySQL)*	Rollback transaction automatically	N/A in MySQL

⭐ 4. Types of Conditions Handled

1. **SQLEXCEPTION –** Any SQL error
2. **SQLWARNING –** Warnings
3. **NOT FOUND –** Cursor fetch reaches end of rows